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Journal:	<i>Current Zoology</i>
Manuscript ID	CZ-2026-0008.R2
Manuscript Type:	Original Article
Date Submitted by the Author:	n/a
Complete List of Authors:	Kelly, Clint; UQAM Departement des sciences biologiques
Keywords:	fecundity, life-history trade off, residual correlation, mixed models, egg size, clutch size

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Disentangling within- and between-female egg size–number trade-offs in the European earwig.

Clint D. Kelly^{1,✉}

¹ Département des Sciences biologiques, Université du Québec à Montréal, Montréal, Canada

✉ Correspondence: [Clint D. Kelly <kelly.clint@uqam.ca>](mailto:kelly.clint@uqam.ca)

For Review Only

Abstract

Life-history theory predicts a trade-off between offspring size and number, yet empirical studies often fail to detect the expected negative association. This discrepancy may occur because phenotypic correlations confuse variation in resource acquisition with variation in allocation strategy. Using repeated measures of female European earwigs (*Forficula auricularia*), I partitioned covariance between egg size and egg number into among- and within-female components using Bayesian mixed models. The raw phenotypic correlation was near zero. However, females that produced more eggs on average laid smaller eggs, revealing a modest negative association among females that aligns with an allocation trade-off. In contrast, there was no detectable within-female covariance once clutch order was accounted for. Egg number repeatability was effectively zero, suggesting that clutch size varies mainly within rather than among females, while egg size was highly repeatable. Both egg number and egg size decreased in second clutches, and within-clutch egg-size variance increased, indicating a reproductive constraint (i.e., fewer functional ovarioles) under a possibly diminished resource state. Together, these findings demonstrate that the size–number trade-off appears to be expressed among females through consistent differences in allocation strategy, whereas clutch size itself remains flexible. By separating hierarchical sources of variation, this study shows that allocation trade-offs can remain hidden at the phenotypic level and that evolutionary constraints act through stable allocation rules rather than through flexible clutch-size adjustments.

Introduction

Life-history theory predicts that organisms must allocate finite resources among competing fitness components. Because energy and time are limited, investment in one trait constrains investment in others (Roff, 1992; Stearns, 1992). A central expression of this principle is the offspring size–number trade-off: for a given reproductive budget, producing larger offspring reduces the number that can be produced (Parker, 1982; Smith and Fretwell, 1974). Since larger eggs often enhance offspring performance and survival (Clutton-Brock, 1991), selection is expected to favour an optimal balance between egg size and egg number that maximises maternal fitness (Fox and Czesak, 2000). Yet empirical tests of the size–number trade-off often fail to reveal the expected negative relationship. Across individuals within populations, egg size and egg number are often positively correlated (Bernardo, 1996; Fox and Czesak, 2000; Hargitai et al., 2005; Roff, 1992). This apparent contradiction arises because phenotypic correlations conflate two distinct biological processes: variation in resource acquisition and variation in allocation strategy. Van Noordwijk and De Jong (1986) showed that when variance in acquisition exceeds variance in allocation, individuals with greater resource pools can produce both more and larger eggs, generating positive among-individual correlations even if all individuals are constrained by the same underlying allocation trade-off. In such cases, the population-level pattern masks allocation constraints operating within individuals—an outcome conceptually analogous to Simpson’s paradox (Simpson, 1951).

From this perspective, the critical prediction of life-history theory is not simply a negative phenotypic correlation, but a negative within-individual covariance between egg size and egg number once among-individual differences in acquisition are accounted for (Careau and Wilson,

2017). Detecting such a masked trade-off, therefore, requires partitioning phenotypic variance and covariance into hierarchical components. Repeated measures of individuals allow estimation of (i) among-individual variance, reflecting consistent differences among females in total reproductive investment, and (ii) within-individual variance, reflecting how females adjust allocation across reproductive bouts. Careau and Wilson (2017) argue that only by decomposing covariance into these components using multivariate mixed models can we determine whether allocation trade-offs operate within individuals despite positive population-level associations. Crucially, this framework also predicts that traits involved in allocation trade-offs should show substantial repeatability. If females differ consistently in resource acquisition, egg size and egg number should exhibit among-individual variance across clutches. High repeatability would, therefore, indicate persistent differences in total reproductive investment among females, consistent with variation in acquisition. The marked increase in within-clutch egg size variance in second broods further supports a resource constraint interpretation. Under reduced resource availability, females may be less able to maintain uniform provisioning across all eggs, leading to greater heterogeneity in egg size within a clutch. Such increased variance is consistent with diminished allocation precision when reproductive resources are limited, rather than an adaptive diversification of offspring size.

Despite the foundational importance of the size–number trade-off, empirical tests rarely partition covariance into within- and among-individual components to distinguish allocation from acquisition effects. Here, I address this gap using the European earwig, *Forficula auricularia*, a species that allows repeated measures of reproductive investment within females but whose pattern of allocation to egg number and size is unclear (Koch and Meunier, 2014). Female

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4 71 earwigs lay a clutch within a short temporal window in a subterranean burrow (Meunier, 2024)
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6 72 and provide extended maternal care (Meunier, 2024; Staerkle and Kölliker, 2008). In Quebec
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8 73 populations, environmental constraints typically render females effectively semelparous
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10 74 (Tourneur, 2018), but under favourable laboratory conditions, they can produce a second clutch.
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12 75 This system, therefore, enables direct estimation of within-female covariance across successive
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14 76 reproductive bouts. I test three predictions derived from acquisition–allocation theory (Van
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16 77 Noordwijk and Jong, 1986; see also Reznick, Nunney, and Tessier, 2000). First, if variance in
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18 78 resource acquisition among females exceeds variance in allocation strategy, then egg size and egg
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20 79 number should exhibit substantial repeatability and a positive among-individual correlation,
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22 80 whereas the within-individual correlation should be negative, revealing a masked allocation
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24 81 trade-off. Second, if maternal fitness is maximised at a particular offspring size or number (Smith
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26 82 and Fretwell, 1974), then one trait should remain relatively invariant across clutches while the
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28 83 other adjusts between reproductive bouts (Winkler and Wallin, 1987). Third, if reproductive
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30 84 investment is structured differently across bouts, then variance in egg size may differ between
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32 85 first and second clutches at both the among-female and within-clutch levels (e.g. Hargitai et al.,
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34 86 2005), providing further insight into how selection shapes the distribution of maternal investment
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36 87 across reproductive events.
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43 88 By separating among-female differences in total reproductive investment from within-female
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45 89 allocation decisions, this study tests a fundamental question in life-history theory: do apparent
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47 90 positive associations between egg size and number reflect relaxed constraints, or do allocation
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49 91 trade-offs persist but remain hidden within individuals? Demonstrating a negative within-female
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51 92 covariance despite positive population-level patterns would show that acquisition can mask, but
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not erase, the constraints imposed by finite resources. In this way, my study evaluates not only the existence of a trade-off but also the hierarchical structure through which life-history constraints are expressed.

Methods

I collected adult female earwigs (n=42) in September and October 2024 in an urban park (Parc Outremont) in Montréal, Canada. All earwigs were brought into the laboratory, assigned a unique ID, and individually placed in a 90-mm-diameter Petri dish containing a thin layer of moist sand. Females were provided a cotton-plugged water vial and fed TetraMin Tropical fish flakes ad libitum. Females were maintained in an incubator (Percival Scientific, Iowa, USA) at 10°C in constant darkness.

Females were checked daily for oviposition. When a female laid eggs, I recorded the date and collected all eggs from the sand and placed them on filter paper in a new Petri dish. The eggs were then photographed under a Leica S6D stereo microscope (Leica Microsystems Inc., Concord, ON, Canada) at 0.68x magnification using Enersight software (Leica Microsystems Inc., Concord, ON, Canada), which automatically adds a scale bar to the photo. Eggs were then discarded.

To stimulate females to lay a second clutch of eggs, I recreated a brief period of spring where they were allowed to accumulate resources for egg production and replenish their sperm stores. To achieve this, females that had recently oviposited were placed in a new Petri dish containing a cotton-plugged water vial, TetraMin Tropical fish flakes, and a shelter consisting of four 4 cm-long straws (Blowholes, Markham, Canada) glued together. These Petri dishes were placed in a

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4 114 23 °C incubator with a 12L:12D light: dark cycle. Females were maintained under these
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6 115 conditions for two weeks, at which time a haphazardly chosen stock male from our colony was
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8 116 placed with her. The pair was kept together for a week, after which a new stock male replaced the
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10 117 original male, and that pair was kept together for another week. Females were then transferred to
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12 118 the 10 °C incubator and treated as previously until they laid their second clutch, at which point
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14 119 their eggs were photographed. Females were euthanised by freezing after laying their second
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16 120 clutch, and their pronotum was photographed as described above. Because female fecundity is
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18 121 often related to body size in insects (Honek, 1993) and could thus possibly explain a positive
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20 122 among-female covariance between egg number and size, I measured two proxies of size,
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22 123 pronotum length and body mass. Females were weighed to the nearest 0.01 mg using a Sartorius
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24 124 Secura 224-1S analytical balance. Females were weighed three times, with each weighing bout
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26 125 separated by four days. I used Fiji image analysis software (Schindelin et al., 2012) to measure
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28 126 the length of the female pronotum to the nearest 0.001 mm three times, with each measurement
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30 127 separated by 4 days. I used the same procedure to measure the perimeter of each egg.
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37 128 **Statistical analysis**

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39 129 I calculated the repeatability of the body mass, pronotum length and egg perimeter measurements
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41 130 using the R package *rptR* (Stoffel, Nakagawa, and Schielzeth, 2017). I entered the trait of interest
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43 131 as the response variable, female ID as a random effect, and assumed a Gaussian distribution.
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45 132 Each analysis was based on 1000 bootstrap samples. The measurements of my two
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47 133 morphological traits were highly repeatable (pronotum length: R [95% CI] = 0.98 [0.97, 0.99];
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49 134 body mass: R [95% CI] = 0.99 [0.988, 0.996]); therefore, I used each female's mean value for
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51 135 each trait where necessary. Because egg perimeter was highly repeatable in both clutches (first
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4 136 clutch: R [95% CI] = 0.977 [0.976, 0.979]; second clutch: R [95% CI] = 0.93 [0.926, 0.937]), I
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6 137 calculated each egg's average perimeter and then used these values to calculate the mean egg size
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8 138 for each clutch for each female.
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11 139 To reduce the two correlated measures of body size (pronotum length and body mass) to a single
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13 140 structural size metric, I performed a principal components analysis (PCA) on the correlation
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15 141 matrix of standardized variables using the *prcomp* function in the R stats package (R Core Team,
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17 142 2025). The first principal component (PC1) explained 60.8% of the total variance and was used
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19 143 as an index of structural body size. In addition, female body condition was estimated using the
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21 144 scaled mass index (SMI) following Peig and Green (2010, 2009; see also Kelly, Tawes, and
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23 145 Worthington, 2014), which standardizes body mass relative to structural size (pronotum length).
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28 146 I first quantified associations among reproductive traits using raw phenotypic correlations.
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30 147 Pearson correlation coefficients were calculated using individual-level trait values, thereby
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32 148 capturing total phenotypic covariation across all sources of variation. These correlations reflect
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34 149 the combined contribution of consistent differences among females, variation among clutches
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36 150 within females, and residual effects. Consequently, raw phenotypic correlations provide a
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38 151 descriptive estimate of overall trait associations but do not distinguish between covariation
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40 152 arising from stable female differences and that arising from within-female (i.e., among-clutch)
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47 154 To distinguish covariance arising from within-female plasticity from covariance attributable to
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49 155 consistent among-female differences, I followed the within-between decomposition approach of
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51 156 van de Pol and Wright [(2009); see Supplementary Material for details]. This decomposition is
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4 157 necessary because residual covariance does not isolate within-individual effects and, in models
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6 158 with different response distributions, does not provide a directly interpretable estimate of within-
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8 159 female trade-offs. Although mean-centering has been shown to impose assumptions about
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10 160 underlying biological processes (Westneat et al., 2020), these assumptions are appropriate here
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12 161 because the biological question concerns how reproductive allocation varies within females
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14 162 across clutches versus how females differ consistently in allocation strategy. Egg number was
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16 163 decomposed into two components: a between-female component, defined as each female’s mean
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18 164 egg number across broods centered on the population mean, and a within-female component,
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20 165 defined as the deviation of each brood from that female’s own mean. Hereafter, we refer to
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22 166 “among-female” variation as differences between individuals and “within-female” variation as
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24 167 differences among clutches within individuals. These two predictors were included
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26 168 simultaneously in a Bayesian multivariate mixed-effects model in which egg number and mean
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28 169 egg size were modelled jointly. Egg number was modelled using a negative binomial distribution
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30 170 with a log link to account for overdispersion, whereas mean egg size was modelled assuming a
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32 171 Gaussian distribution with an identity link. Clutch (first vs. second clutch) was included as a
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34 172 fixed effect in both submodels. Female identity was included as a random intercept for both
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36 173 response variables. This structure allowed estimation of among-female variance in egg number,
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38 174 among-female variance in egg size, and the covariance (correlation) between female-level
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40 175 intercepts for the two traits.
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48 176 This multivariate framework partitions total phenotypic covariance into components arising from
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50 177 consistent differences among females and from within-female deviations across clutches.
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52 178 Covariance attributable to consistent differences among females is captured by the female-level
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random effects, whereas covariance arising from within-female variation is captured by the fixed within-female component of egg number. The fixed between-female component estimates how mean egg size varies with mean egg number across females, and thus captures systematic (i.e., explained) differences in allocation strategy among females. In contrast, the fixed within-female component tests whether egg size changes plastically when a female produces more or fewer eggs than her own mean. The correlation between female-level intercepts represents residual (latent) covariance among baseline trait values that is not explained by the fixed effects, including the between-female allocation relationship. Thus, the between-female slope tests whether females diverge in their mean allocation strategy (i.e., whether females that consistently produce more eggs also produce smaller eggs), whereas the intercept correlation quantifies any remaining covariance between egg number and egg size after accounting for these effects. Continuous predictors were standardized prior to analysis, such that standardized coefficients represent the expected change in egg size (in standard deviation units) associated with a one standard deviation change in egg number at the relevant hierarchical level.

To quantify the structure of egg-size variation specifically, I fitted a univariate mixed-effects model in which egg size (standardised mean egg perimeter) was modelled as a Gaussian response with female identity included as a random intercept (see Supplementary Material for details). This formulation partitions total phenotypic variance in egg size into variance attributable to differences among females and variance occurring within females among eggs within clutches. The among-female variance component reflects consistent maternal differences in egg size allocation, whereas the within-female component captures heterogeneity in provisioning within clutches and residual variation.

Repeatability was calculated on the observation scale for both traits. For egg size (Gaussian model):

$$R_{\text{egg size}} = \frac{V_F}{V_F + V_R}$$

where V_F is the among-female variance (i.e., variance of the female random intercept) and V_R is the residual variance. Because the Gaussian model is parameterized directly on the observation scale, this formulation corresponds to both latent- and observation-scale repeatability.

For egg number, which was modelled using a negative binomial distribution with a log link, the variance depends on both the expected mean and the dispersion (shape) parameter. Under the negative binomial parameterization used in *brms*, the observation-level (distribution-specific) variance is given by:

$$\text{Var}(Y) = \mu + \frac{\mu^2}{\theta}$$

where μ is the expected mean egg number and θ is the dispersion (shape) parameter. Because the model includes a log link, the among-female variance estimated from the random intercept is expressed on the latent (log) scale and must be transformed to the observation scale before calculating repeatability. Following Nakagawa et al. (2017), the among-female variance on the observation scale was therefore calculated as:

$$V_F^{\text{obs}} = \mu^2(\exp(\sigma^2) - 1)$$

where σ^2 is the variance of the random intercept on the latent scale.

Observation-scale repeatability for egg number was then calculated as:

$$R_{\text{egg number}} = \frac{V_F^{\text{obs}}}{V_F^{\text{obs}} + \left(\mu + \frac{\mu^2}{\theta}\right)}$$

Both the transformed among-female variance and the distribution-specific variance were computed for each posterior draw using the corresponding posterior samples of μ , σ^2 , and θ . To account for variation in expected values across observations, these quantities were first calculated at the observation level and then averaged within each posterior draw. This procedure yielded a posterior distribution of repeatability that fully propagates uncertainty from all variance components.

Repeatability for non-Gaussian traits was calculated following the framework of Nakagawa and Schielzeth (2010) and its extensions for generalized linear mixed models (Nakagawa, Johnson, and Schielzeth, 2017). Because distribution-specific variance can contribute substantially to total phenotypic variance in negative binomial models, repeatability estimates for count data are often lower than those obtained under Gaussian assumptions, even when among-individual variance is present (Nakagawa, Johnson, and Schielzeth, 2017; Nakagawa and Schielzeth, 2010). By calculating repeatability on the observation scale, these estimates represent the proportion of total phenotypic variance in observed egg number attributable to consistent differences among females.

I used weakly informative priors to constrain parameters to biologically realistic ranges while allowing the data to drive inference. Regression coefficients were assigned normal priors centered at zero (Normal(0, 0.5–1)), reflecting the expectation that clutch effects on egg traits would be moderate in magnitude. Intercepts were given normal priors scaled to each trait (egg

number: Normal(0, 5); mean egg size: Normal(3.5, 1); standardized egg size: Normal(0, 0.5)). Among-female variation was assigned exponential priors (Exponential(1–2)), which favour small but non-zero individual differences. Residual variance (Gaussian models) and the negative binomial dispersion parameter were given Exponential(1) priors. Correlations among female-level random effects were regularized using an LKJ(2) prior to avoid overestimating trait associations. Posterior distributions were obtained using six Markov chains with 4,000 iterations per chain, including 1,000 warm-up iterations, and convergence was assessed using $\hat{R}$ statistics, effective sample sizes, and trace-plot inspection. Posterior distributions are summarised as posterior means with 95% credible intervals. Data were visualized using *ggplot* (Wickham, 2016) and *tidybayes* (Kay, 2020).

Results

Between- and within-female associations

Across all broods, the raw phenotypic correlation between egg number and egg size was weak and near zero ($r_{raw} = -0.04$, 95% CI [-0.25, 0.19]), indicating no detectable population-level association when hierarchical structure was ignored.

Egg number (clutch one: $\bar{x} = 51.55 \pm 2.49$ eggs; clutch two: $\bar{x} = 34.14 \pm 2.65$ eggs) and egg size (first clutch: $\bar{x} = 3.54 \pm 0.03$ mm; second clutch: $\bar{x} \pm SE = 3.41 \pm 0.03$ mm) both declined in second clutches. Egg number decreased on the log scale ($\beta = -0.41$, 95% CrI (-0.64, -0.19; [Figure 1](#); Supplementary Table 1), corresponding to an approximate 34% reduction on the natural scale ($e^{-0.413} \approx 0.66$). Mean egg size also declined in the second clutch ($\beta = -0.13$, 95% CrI (-0.24, -0.02)).

The results did not support my prediction of a within-female plastic trade-off between egg number and egg size (Figure 1 a). The within-female deviation in egg number had a slope near zero ($\beta_W = -0.01$, 95% CrI (-0.06, 0.05), indicating that females did not systematically adjust egg size when producing more or fewer eggs relative to their own mean. By comparison, the among-female association (captured by the fixed between-female slope) revealed a negative relationship between egg size and mean female egg number ($\beta_B = -0.05$, 95% CrI (-0.10, 0.00; Figure 1 b). The posterior distribution of the among-female slope was centered on negative values, suggesting weak evidence for divergence in mean allocation strategy among females. The correlation between female-level intercepts for egg number and egg size was highly uncertain (median = -0.02, 95% CrI: -0.81, 0.82), indicating limited information for estimating latent covariance with only two broods per female. With only two repeated measures per female, intercept correlations are weakly identifiable and should be interpreted cautiously.

No measure of female size/condition was positively correlated with the number of eggs in the first brood (pronotum length: $r = 0.14$, 95% CI [-0.19, 0.43]; body mass: $r = 0.12$, 95% CI [-0.21, 0.43]; PC1: $r = -0.17$, 95% CI [-0.44, 0.15]; scaled mass index: $r_{raw} = -0.02$, 95% CI [-0.35, 0.33]) providing no evidence that larger females or those in better condition produce more eggs. I therefore did not include a measure of size/condition in this model. Together, these results show that the size–number trade-off is not apparent at the raw phenotypic level and is not expressed as within-female plasticity, but emerges as a modest negative association at the among-female level through differences in average allocation strategy.

Egg size variation between females and within clutches

Egg size exhibited substantial variation at both the between-female and within-clutch levels (Figure 2; Supplementary Table 2). Between-female variance was similar across reproductive events, with a median of 0.73 (95% CrI: 0.50, 1.12) in first clutches and 0.61 (95% CrI: 0.41, 0.96) in second clutches. The posterior median difference (second – first) was -0.11 (95% CrI: -0.56, 0.28), and the posterior probability that between-female variance was greater in second clutches was 0.28, indicating no clear evidence of a difference. In contrast, within-clutch (egg-to-egg) variance was substantially higher in second clutches (median = 0.35 (95% CrI: 0.33, 0.38) than in first clutches (median = 0.21 (95% CrI: 0.20, 0.22). The posterior median difference was 0.14 (95% CrI: 0.12, 0.17), with a posterior probability of 1.00, that within-clutch variance was greater in second clutches.

Egg number and egg size repeatability

Repeatability was calculated from posterior samples of the variance components on the observation scale. Egg number exhibited extremely low repeatability. Although there was some among-female variance in egg number (SD = 0.078, 95% CrI [0.003, 0.270]; Supplementary Table 1), the overwhelming majority of variance arose from distribution-specific (negative binomial) variance. Consequently, observation-scale repeatability was near zero ($R = 4.86 \times 10^{-2}$, 95% CrI [3.80×10^{-5} , 2.43×10^{-1}]), indicating that very little of the total phenotypic variation in clutch size was attributable to consistent differences among females.

In contrast, egg size showed moderate repeatability on the observation scale. Among-female variance (SD = 0.082, 95% CrI [0.006, 0.155]; Supplementary Table 1) was substantial relative to residual variance (SD = 0.191, 95% CrI [0.155, 0.235]), yielding $R = 0.78$ (95% CrI: 0.70,

0.84) in first broods and $R = 0.63$ (95% CrI: 0.53, 0.73) in second broods. These results indicate that females differed consistently in their mean egg size across broods, whereas clutch size was largely context-dependent.

Discussion

Life-history theory predicts a trade-off between offspring size and number because both traits draw from a finite pool of reproductive resources (Lack, 1947; Roff, 1992; Smith and Fretwell, 1974; Stearns, 1992). Yet, as emphasised by van Noordwijk and de Jong (1992), phenotypic correlations need not directly reveal allocation trade-offs, because variation in resource acquisition can obscure underlying allocation rules (Careau and Wilson, 2017; Johnson and Nassrullah, 2024). This acquisition–allocation framework has become central to interpreting life-history covariance (Houle, 1991; Reznick, Nunney, and Tessier, 2000). By partitioning covariance into within- and among-female components, my results show that the size–number trade-off in this population of earwigs is at most weakly expressed among females and absent within them, illustrating how hierarchical structure shapes the detectability of life-history trade-offs (Careau and Wilson, 2017; Dingemanse and Dochtermann, 2013; Pol and Wright, 2009; Westneat et al., 2020).

At the phenotypic level, egg size and egg number were uncorrelated in my earwig population, consistent with Koch and Meunier (2014). Under the framework of van Noordwijk and de Jong (1986), such phenotypic independence is expected when variation in resource acquisition obscures underlying allocation trade-offs. After hierarchical decomposition, a weak negative association emerged at the among-female level: females that produced more eggs on average

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4 324 tended to lay slightly smaller eggs. This pattern is captured by the fixed between-female slope
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6 325 and reflects systematic differences among females in their mean allocation strategy. In contrast,
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8 326 the correlation between female-level intercepts—representing residual (latent) covariance after
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10 327 accounting for this allocation relationship—was highly uncertain and overlapped zero, indicating
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12 328 little evidence for additional integration between traits beyond that explained by differences in
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15 329 mean allocation strategy (Roff and Fairbairn, 2007; Stearns, 1989).
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18 330 Crucially, these results must be interpreted in light of the variance structure of egg number. Egg
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20 331 number exhibited near-zero repeatability, indicating that there is very little residual among-
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22 332 female variation in clutch size once distribution-level variance is taken into account. At first
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24 333 glance, this may appear inconsistent with the presence of an among-female association. However,
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26 334 repeatability and the between-female slope quantify different components of variation.
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28 335 Repeatability reflects the proportion of unexplained (random-effect) variance attributable to
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30 336 consistent differences among females, whereas the between-female slope represents a systematic
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32 337 (fixed-effect) relationship among female means. Thus, even when residual among-female
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34 338 variance is very small, it is still possible to detect a weak association among female means. In
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36 339 this case, the near-zero repeatability indicates that clutch size does not vary consistently among
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38 340 females, while the between-female slope suggests only minor differences in average reproductive
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40 341 output that covary weakly with egg size. Taken together, these results indicate that the apparent
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42 342 among-female trade-off is weak and arises from small differences in female means rather than
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44 343 from substantial or repeatable variation in clutch size. Accordingly, there is limited evidence for
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46 344 stable divergence in allocation strategy among females.
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4 345 In contrast, there was no evidence for a within-female plastic trade-off. Although each female
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6 346 contributed only two broods, the posterior distribution of the within-female slope was tightly
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8 347 centered on zero, suggesting that any within-female effect, if present, is small. Females did not
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10 348 reduce egg size when producing more eggs than their own average; instead, egg number and egg
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12 349 size shifted together across broods. This pattern is consistent with state-dependent life-history
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14 350 models in which allocation rules remain relatively stable while total resource state fluctuates
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16 351 across reproductive bouts (Houston and McNamara, 1999; McNamara and Houston, 1996).
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18 352 Rather than dynamically reallocating resources between offspring size and number, females
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20 353 appear to exhibit clutch-level shifts driven by overall energetic state. Together, these results
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22 354 suggest that clutch size is primarily state-dependent and influenced by transient physiological or
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24 355 structural constraints, whereas egg size reflects a more stable component of reproductive
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26 356 allocation.
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32 357 Notably, egg number was not correlated with female structural size or body condition (i.e. scaled
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34 358 mass), indicating that persistent differences in morphology do not explain clutch-size variation. If
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36 359 stable acquisition differences were driving among-female variation, we would expect egg number
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38 360 to show repeatability and to covary with these morphological size/condition indicators. The
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40 361 absence of such patterns weakens the interpretation that persistent acquisition differences—at
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42 362 least as reflected by morphology—underlie among-female differences in reproductive output.
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44 363 Instead, clutch size appears predominantly state-sensitive and influenced by clutch-level
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46 364 constraints.
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51 365 This interpretation also helps explain the pronounced decline in second clutches. That second
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53 366 clutches comprised both fewer and smaller eggs is consistent with previous work in North
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4 367 American and European populations (Koch and Meunier, 2014; Tourneur and Gingras, 1992).
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6 368 Tourneur and Gingras (1992) proposed that some ovarioles become non-functional after the first
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8 369 clutch, thereby reducing egg production capacity. Structural limitation could therefore contribute
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10 370 to smaller second clutches. However, structural reduction alone cannot explain the simultaneous
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12 371 decline in egg size. Under a strict allocation trade-off, a reduction in egg number would be
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14 372 expected to lead to compensatory increases in egg size if total resources remain constant. Instead,
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16 373 both traits declined, indicating that females likely entered the second reproductive bout with
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18 374 diminished resource reserves in addition to any physiological constraint. In the wild, females
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20 375 accumulate resources for months before producing their first clutch, whereas in the laboratory,
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22 376 they had only a brief interval (approx. one month) to replenish reserves before producing the
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24 377 second. Late reproduction, therefore, likely reflects both reduced energetic state and structural
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26 378 limitation.
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31
32 379 The hierarchical variance structure supports a constraint-based interpretation of clutch-size
33
34 380 variation. Egg size exhibited moderate repeatability across broods, indicating stable differences
35
36 381 among females in mean per-offspring investment. In contrast, egg number showed extremely low
37
38 382 repeatability, with most variance arising from brood effects and distribution-level variance rather
39
40 383 than persistent female differences. The absence of any association between egg number and
41
42 384 female structural size or condition, together with the near-zero repeatability of clutch size,
43
44 385 suggests that variation in egg number is not driven by stable differences in resource acquisition
45
46 386 but instead reflects transient physiological state or clutch-level constraint. Thus, clutch size
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48 387 appears primarily state-sensitive, whereas egg size reflects a more stable allocation strategy
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52 388 (Houston and McNamara, 1999; McNamara and Houston, 1996). At the same time, within-clutch
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variance increased substantially in second broods, indicating reduced developmental precision under diminished resource availability. Structural and energetic constraints, therefore, appear to reduce total reproductive output while leaving mean allocation rules largely intact (Ricklefs and Wikelski, 2002; Zera and Harshman, 2001). These patterns have broader implications for phenotypic integration and evolutionary potential. Repeatability places an upper bound on heritability (Falconer and MacKay, 1996; Lessells and Boag, 1987; Nakagawa and Schielzeth, 2013; Nakagawa and Schielzeth, 2010; but see Dohm, 2002). The moderate repeatability of egg size suggests that selection could act on persistent differences in mean provisioning, whereas the extremely low repeatability of egg number implies that clutch size may respond primarily to short-term environmental or physiological conditions (i.e. reduced ovariole function). The weak intercept correlation further indicates that egg size and egg number are only loosely integrated, consistent with models in which traits differ in their sensitivity to resource state versus allocation strategy (Houle, 1991). In this system, per-offspring investment appears more evolutionarily stable than total offspring number.

Viewed hierarchically, my results illustrate a broader conceptual point: life-history trade-offs may operate primarily through systematic differences in mean allocation among individuals, while most short-term variation reflects state-dependent constraint. However, when among-individual variance in one trait is minimal—as is the case for egg number here—such trade-offs will necessarily be weak and difficult to detect. Without decomposing within- and among-individual covariance, even this modest pattern would have remained obscured. More generally, the evolutionary consequences of trade-offs depend not only on their sign but on their hierarchical expression and repeatability. A trade-off that reflects consistent among-individual

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4 411 differences in a repeatable trait is more likely to shape evolutionary responses than one arising
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6 412 primarily from transient, plastic variation within individuals. Future work measuring resource
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8 413 acquisition directly, experimentally manipulating body condition, or estimating genetic
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10 414 correlations between egg size and egg number will help determine whether the limited allocation
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12 415 divergence observed here reflects adaptive differentiation or condition-dependent constraint
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15 416 (Lande, 1980; Roff, 1992).
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19

20 417 **Acknowledgements**

21
22 418 I thank Andréanne Nault and Clara Smythe for assistance in animal care and data collection. I
23
24 419 also thank Denis Réale, Pierre-Olivier Montiglio, and two anonymous reviewers for their
25
26 420 discussion and feedback.
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30

31 421 **Funding**

32
33 422 The study was funded by a Discovery Grant from the Natural Sciences and Engineering Research
34
35 423 Council of Canada (NSERC).
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40 424 **Conflict of Interest**

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42 425 I declare no conflict of interest.
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46 426 **Data availability**

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48 427 Data are available through the OSF repository at 10.17605/OSF.IO/KWCS8.
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Author contributions

CDK collected and analyzed the data, and wrote the manuscript.

Ethics Statement

All data were collected using the European earwig (*Forficula auricularia*), an invertebrate species for which institutional animal care approval and collection permits are not required under current regulations. All procedures adhered to standard ethical guidelines for the care and use of animals in research.

Artificial intelligence (AI)-assisted technologies

Alt text for the two figures was generated with the assistance of AI tools and reviewed and verified by the author for accuracy and completeness.

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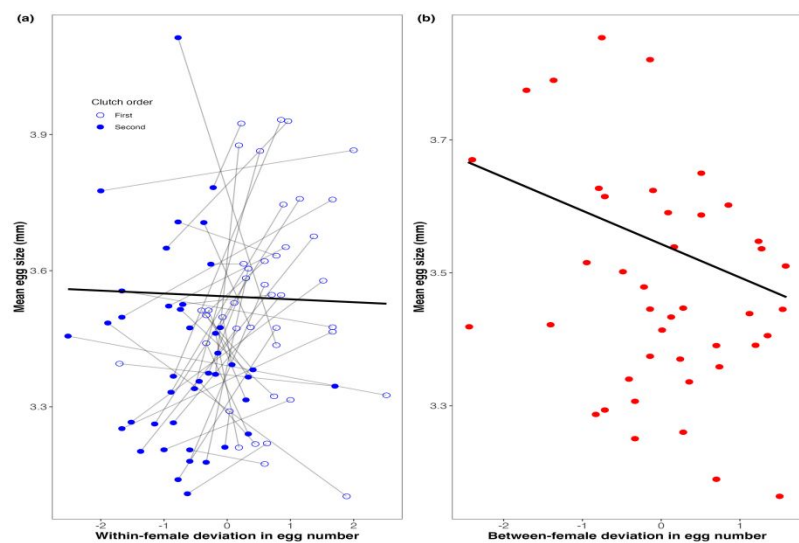
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1
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4 538 ##Figures



25
26 Figure 1: Partitioning the egg size–number trade-off into within- and between-female
27 components. (a) Within-female (plastic) relationship between egg number and mean egg size.
28 Points represent individual broods (first clutch: open circles; second clutch: closed circles),
29 with grey lines connecting repeated measures from the same female. The x-axis shows
30 standardized within-female deviation in egg number (0 = female mean). The solid black line
31 represents the Bayesian population-level estimate of the within-female slope ($\beta_W = -0.01$, 95%
32 CrI (-0.06, 0.05)). (b) Between-female relationship (i.e., differences in mean allocation strategy)
33 based on female means across broods plotted against standardized between-female deviation in
34 egg number. The black line represents the Bayesian population-level between-female slope (β_B
35 = -0.05, 95% CrI (-0.10, 0.00)). The panels distinguish plastic allocation within females from
36 strategic differences among females, indicating weak support for a reproductive size–number
37 trade-off.
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41 539 Alt text: Two-panel scatterplot showing relationships between egg number and mean egg size.
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43 540 Panel (a) plots within-female deviations, with paired points for first and second clutches
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45 541 connected by lines; the trend line is nearly flat, indicating little within-female trade-off. Panel (b)
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47 542 plots between-female deviations, with a downward-sloping regression line indicating that females
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49 543 producing more eggs tend to have smaller eggs.
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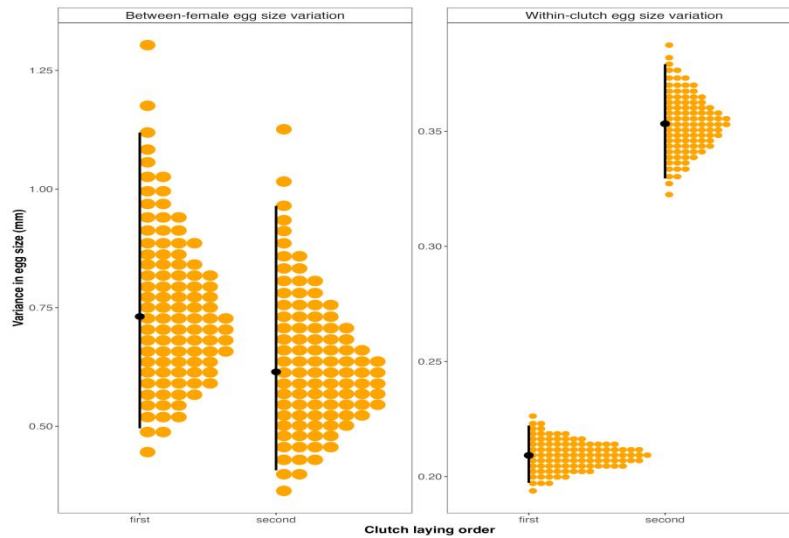
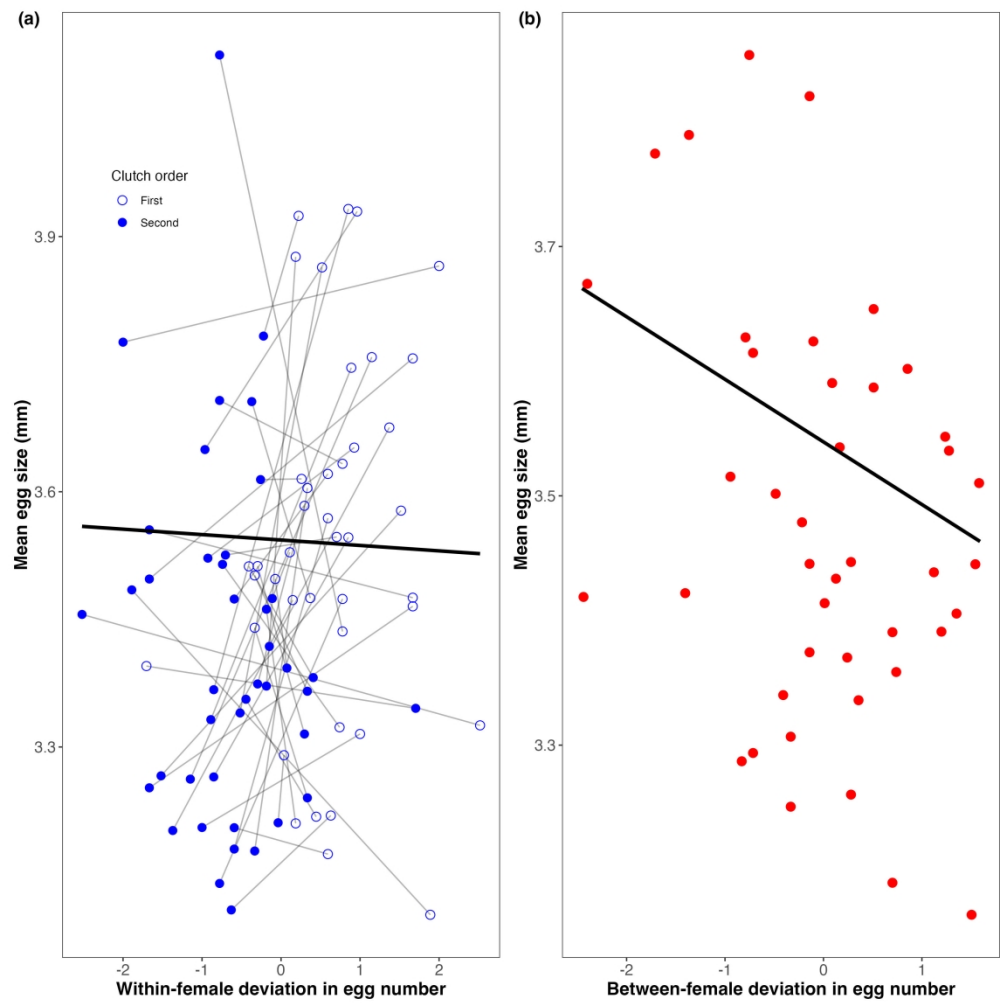


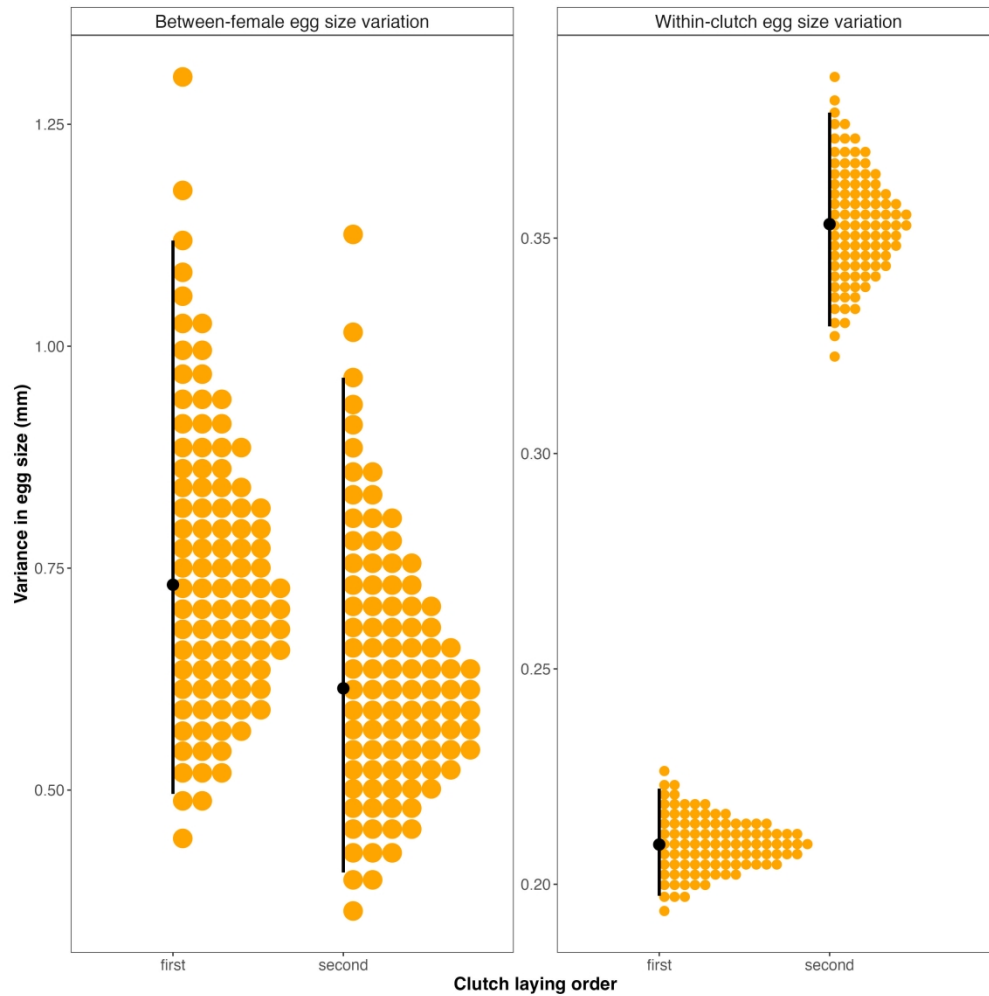
Figure 2: Clutch-specific variance components of egg size. (a) Between-female variance in egg size for first and second clutches, estimated from a hierarchical Bayesian model with clutch-specific female effects. (b) Within-clutch (egg-to-egg) variance for first and second clutches. Orange points represent posterior draws, black points indicate posterior medians, and vertical lines denote 95% credible intervals. Between-female variance did not differ clearly between clutches, whereas within-clutch variance was greater in second clutches.

Alt text: Two-panel dot plot showing variation in egg size across clutch order. The left panel shows between-female variation, with distributions for first and second clutches; variance is higher for first clutches than second clutches, with mean values indicated by black points and vertical error bars. The right panel shows within-clutch variation, with second clutches exhibiting greater variance than first clutches; means and error bars are shown in black.



Partitioning the egg size–number trade-off into within- and between-female components. (a) Within-female (plastic) relationship between egg number and mean egg size. Points represent individual broods (first clutch: open circles; second clutch: closed circles), with grey lines connecting repeated measures from the same female. The x-axis shows standardized within-female deviation in egg number (0 = female mean). The solid black line represents the Bayesian population-level estimate of the within-female slope ($\beta_W = -0.01$, 95% CrI (-0.06, 0.05)). (b) Between-female relationship (i.e., differences in mean allocation strategy) based on female means across broods plotted against standardized between-female deviation in egg number. The black line represents the Bayesian population-level between-female slope ($\beta_B = -0.05$, 95% CrI (-0.10, 0.00)). Together, the panels distinguish plastic allocation within females from strategic differences among females, indicating weak support for a reproductive size–number trade-off.

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Clutch-specific variance components of egg size. (a) Between-female variance in egg size for first and second clutches, estimated from a hierarchical Bayesian model with clutch-specific female effects. (b) Within-clutch (egg-to-egg) variance for first and second clutches. Grey points represent posterior draws, black points indicate posterior medians, and vertical lines denote 95% credible intervals. Between-female variance did not differ clearly between clutches, whereas within-clutch variance was greater in second clutches.

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**Supplemental Material: Disentangling within- and between-
female egg size–number trade-offs in the European earwig**

Clint D. Kelly^{1,✉}

¹ Département des Sciences biologiques, Université du Québec à Montréal, Montréal, Canada

✉ Correspondence: [Clint D. Kelly <kelly.clint@uqam.ca>](mailto:kelly.clint@uqam.ca)

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1. Within–between multivariate model

To partition covariance arising from within-female plasticity from covariance attributable to consistent among-female differences, I applied the within–between decomposition of van de Pol and Wright (Pol and Wright, 2009). Although mean-centering can impose assumptions about underlying biological processes (Westneat et al., 2020), these assumptions are appropriate here because the aim was to distinguish plastic allocation within females from persistent differences among females.

Let egg number for female i in brood j be denoted N_{ij} . Egg number was decomposed as

$$N_{ij} = \bar{N}_i + (N_{ij} - \bar{N}_i) \quad (1)$$

The between- and within-female components were defined as

$$N_i^B = \bar{N}_i - \bar{N} \quad (2)$$

$$N_{ij}^W = N_{ij} - \bar{N}_i \quad (3)$$

Egg number submodel

Egg number was modelled using a negative binomial distribution with a log link:

$$N_{ij} \sim \text{NegBinomial}(\mu_{ij}, \theta) \quad (4)$$

$$\log(\mu_{ij}) = \beta_0^N + \beta_1^N \text{Brood}_{ij} + u_{0i}^N \quad (5)$$

where

$$u_{0i}^N \sim \mathcal{N}(0, \sigma_N^2) \quad (6)$$

Mean egg size submodel

Mean egg size was modelled as

$$S_{ij} \sim \mathcal{N}(\mu_{ij}^S, \sigma_S^2) \tag{7}$$

$$\mu_{ij}^S = \beta_0^S + \beta_1^S \text{Brood}_{ij} + \beta_W N_{ij}^W + \beta_B N_i^B + u_{0i}^S \tag{8}$$

with

$$u_{0i}^S \sim \mathcal{N}(0, \sigma_{S,id}^2) \tag{9}$$

Multivariate structure

The two submodels were fit jointly:

$$\begin{pmatrix} u_{0i}^N \\ u_{0i}^S \end{pmatrix} \sim \mathcal{MVN} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma \right) \tag{10}$$

where

$$\Sigma = \begin{pmatrix} \sigma_N^2 & \sigma_{NS} \\ \sigma_{NS} & \sigma_{S,id}^2 \end{pmatrix} \tag{11}$$

The intercept correlation is

$$\rho_{NS} = \frac{\sigma_{NS}}{\sqrt{\sigma_N^2 \sigma_{S,id}^2}} \tag{12}$$

Relationship between model parameters and total phenotypic covariance

Total phenotypic covariance can be partitioned as

$$\text{Cov}_{\text{total}} = \text{Cov}_{\text{within}} + \text{Cov}_{\text{among}} \tag{13}$$

Within-female covariance

$$\text{Cov}_{\text{within}} = \beta_W \text{Var}(N_{ij}^W) \quad (14)$$

Among-female covariance

$$\text{Cov}_{\text{among, slope}} = \beta_B \text{Var}(N_i^B) \quad (15)$$

$$\text{Cov}_{\text{among, intercept}} = \sigma_{NS} \quad (16)$$

Total among-female covariance is therefore

$$\text{Cov}_{\text{among}} = \beta_B \text{Var}(N_i^B) + \sigma_{NS} \quad (17)$$

Total phenotypic covariance

Combining components:

$$\text{Cov}_{\text{total}} = \beta_W \text{Var}(N_{ij}^W) + \beta_B \text{Var}(N_i^B) + \sigma_{NS} \quad (18)$$

The intercept correlation is

$$\rho_{NS} = \frac{\sigma_{NS}}{\sqrt{\sigma_N^2 \sigma_{S,id}^2}} \quad (19)$$

Corresponding *brms* Specification

```
bf_number <- bf(
  egg.number ~ brood + (1 |p| id),
  family = negbinomial()
)

bf_size <- bf(
  mean.egg.size ~ brood +
    egg_number_within_z +
    egg_number_between_z +
    (1 |p| id),
  family = gaussian()
)
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```
mv_model <- bf_number + bf_size
```

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Table S 1: Posterior summaries from the multivariate Bayesian mixed-effects model examining the relationship between egg number and mean egg size. The model jointly estimated a negative binomial model for egg number and a Gaussian model for egg size, including brood effects and partitioning egg number into within- and between-female components. Reported values are posterior medians and 95% credible intervals (CrI). Random effects represent among-female (ID-level) variation, and the correlation term describes the association between female intercepts for egg number and egg size. The dispersion parameter refers to the negative binomial shape parameter.

Parameter	Estimate	Lower 95% CrI	Upper 95% CrI
Egg number: Brood 2 – Brood 1 (log scale)	-0.413	-0.635	-0.189
Egg number: Intercept (Brood 1, log scale)	3.940	3.782	4.107
Egg size: Brood 2 – Brood 1	-0.133	-0.244	-0.024
Egg size: Between-female deviation in egg number (z)	-0.050	-0.104	0.002
Egg size: Within-female deviation in egg number (z)	-0.007	-0.061	0.049
Egg size: Intercept (Brood 1)	3.543	3.468	3.618
cor_id__eggnumber_Intercept__meaneggsize_Intercept	-0.017	-0.813	0.816
SD (Egg number Female ID)	0.078	0.003	0.270
SD (Egg size Female ID)	0.082	0.006	0.155
Residual SD (Egg size)	0.191	0.155	0.235
Dispersion (Egg number NegBin shape)	4.080	2.835	5.853

2. Clutch-Specific Hierarchical Model of Egg Size

Statistical Model

Egg perimeter was standardized prior to analysis (mean = 0, SD = 1).

For female i , egg j , and brood $b \in \{1,2\}$, standardized egg size y_{ijb} was modeled as:

$$y_{ijb} \sim \mathcal{N}(\mu_{ijb}, \sigma_b^2)$$

The conditional mean structure was:

$$\mu_{ijb} = \beta_0 + \beta_1 I(b = 2) + u_{ib}$$

where:

- β_0 is the mean egg size in clutch 1 (reference level),
- β_1 is the fixed difference between clutch 2 and clutch 1,
- u_{ib} is the female-specific deviation for clutch b .

Random Effects Structure

Clutch-specific female deviations were modeled as a correlated bivariate normal vector:

$$\mathbf{u}_i = \begin{pmatrix} u_{i1} \\ u_{i2} \end{pmatrix} \sim \mathcal{MVN} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma \right)$$

with variance–covariance matrix:

$$\Sigma = \begin{pmatrix} \tau_1^2 & \rho\tau_1\tau_2 \\ \rho\tau_1\tau_2 & \tau_2^2 \end{pmatrix}$$

Here:

- τ_1^2 and τ_2^2 are the among-female variances in clutches 1 and 2, respectively,
- $\rho = \text{Corr}(u_{i1}, u_{i2})$ is the correlation between female deviations across clutches.

Importantly, no clutch-independent random intercept was estimated. The model includes only clutch-specific female deviations (i.e., random slopes of clutch within female), corresponding to the specification `(0 + brood_factor | id)` in *brms*.

Residual Variance Model

Residual variance was allowed to differ between clutches:

$$\log(\sigma_b) = \alpha_0 + \alpha_1 I(b = 2)$$

where:

- σ_1^2 is the residual variance in clutch 1,
- σ_2^2 is the residual variance in clutch 2.

The log link ensures that residual standard deviations remain positive.

Corresponding *brms* Specification

```
bf(
  scale(mean.perim) ~ brood_factor + (0 + brood_factor | id),
  sigma ~ brood_factor
)
```

Table S 2: Posterior medians and 95% credible intervals from a Bayesian mixed model of scaled egg perimeter, including clutch-specific means, clutch-specific female-level random effects, and clutch-specific residual variance.

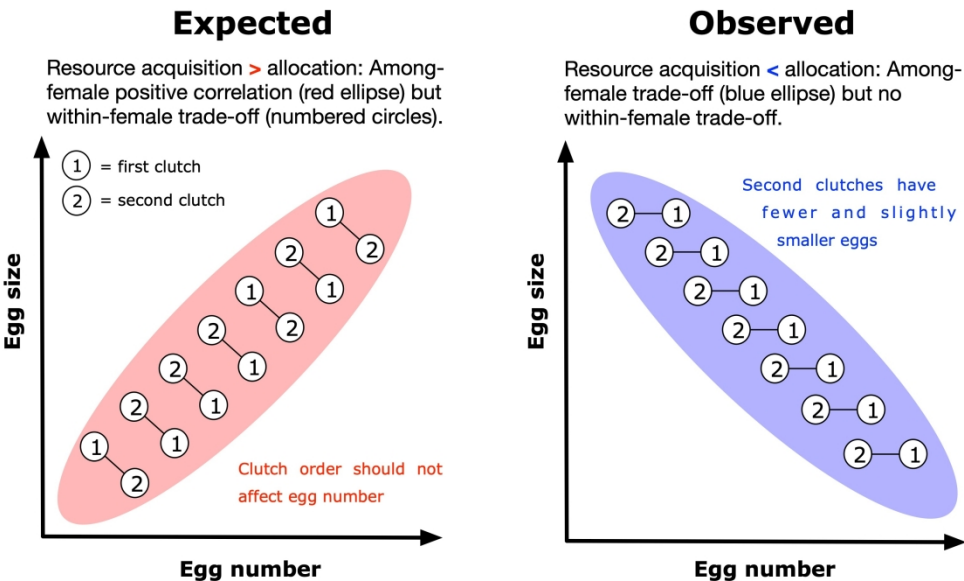
Parameter	Estimate	Lower 95% CrI	Upper 95% CrI
Brood 2 – Brood 1	-0.535	-0.827	-0.246
Mean (Brood 1)	0.283	0.050	0.530
Correlation (Brood 1, Brood 2 Female ID)	0.250	-0.025	0.488
SD (Brood 1 Female ID)	0.855	0.704	1.058
SD (Brood 2 Female ID)	0.784	0.638	0.982
Residual SD (Brood 1)	0.457	0.444	0.471
Residual SD (Brood 2)	0.594	0.574	0.616

References

Pol, M. van de, Wright, J., 2009. A simple method for distinguishing within- versus between-subject effects using mixed models. *Anim Behav* **77**, 753–758.

Westneat, D.F., Araya-Ajoy, Y.G., Allegue, H., Class, B., Dingemanse, N., Dochtermann, N.A., Garamszegi, L.Z., Martin, J.G.A., Nakagawa, S., Réale, D., Schielzeth, H., 2020. Collision between biological process and statistical analysis revealed by mean centring. *J Anim Ecol* **89**, 2813–2824.

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Two-panel conceptual diagram comparing expected versus observed relationships between egg number and egg size. In the left panel (Expected), a positive among-female correlation is shown (red ellipse), with numbered points indicating first and second clutches and a within-female trade-off implied. In the right panel (Observed), a negative among-female relationship is shown (blue ellipse), indicating a trade-off between egg number and size across females, but no within-female trade-off; second clutches are depicted as having fewer and slightly smaller eggs.

268x155mm (300 x 300 DPI)

Dear Dr Huang,

Thank you for the opportunity to revise and resubmit our manuscript, *"Disentangling within- and between-female egg size–number trade-offs in the European earwig."* I am grateful to you and the reviewers for the thoughtful and detailed feedback, which has substantially improved the clarity, accuracy, and interpretation of my study.

I have carefully addressed all comments from both reviewers and made the corresponding revisions throughout the manuscript. Below, I provide a response to each comment, summarised in bullet points.

Clint Kelly

Reviewer: 1

Recommendation: Minor Revision

Comments:

The revised manuscript is substantially improved. The statistical overhaul is genuine and goes well beyond surface-level fixes.

However, the Results section contains numerical errors in the repeatability paragraph (swapped values, wrong table citation, and a scale mixing error in the egg number repeatability computation), some interpretive claims remain stronger than the posteriors warrant, and the OSF data repository remains empty.

MAJOR CONCERNS

1. Numerical Errors in Egg Size Repeatability Reporting

The Results paragraph on egg size repeatability contains two distinct errors:

- a) The variance components cited are from the wrong supplementary table. The text states: "Among-female variance (SD = 0.082, 95% CrI [0.006, 0.155]; Supplementary Table 2) was substantial relative to residual variance (SD = 0.191, 95% CrI [0.155, 0.235])." These values (SD = 0.082, SD = 0.191) appear in Supplementary Table S1 (the multivariate within/between trade-off model), not Table S2 (the clutch-specific egg size model). Table S2 reports clutch-specific among-female SDs of 0.855 (Brood 1) and 0.784 (Brood 2), with residual SDs of 0.457 and 0.594. Computing repeatability from the cited Table S1 values yields $R = 0.082^2 / (0.082^2 + 0.191^2) \sim 0.16$, which does not match either reported repeatability, confirming the values are from the wrong model.

- Supplementary Table 1 should have been cited for the among and residual variances. These values show that among-female variance in egg size is tiny compared to within-female variation. Egg size repeatability was calculated using the clutch-specific values in Supplementary Table 2.

b) The repeatability values are assigned to the wrong clutches. The text reports "R = 0.63 in first broods and R = 0.78 in second broods." However, computing repeatability from the correct Table S2 variance components:

Brood 1: $R = 0.855^2 / (0.855^2 + 0.457^2) = 0.731 / 0.940 = 0.78$
Brood 2: $R = 0.784^2 / (0.784^2 + 0.594^2) = 0.615 / 0.968 = 0.64$ (i.e. the 0.63 reported in the text)

First broods have higher repeatability ($R \sim 0.78$), not second broods. The reported values are transposed.

- Many thanks for pointing that out. That was a Quarto scripting error.

Action required:

- a) Cite the correct variance components from Table S2 (or specify the model from which they derive).
- b) Correct the assignment of repeatability values to the appropriate clutch.

MODERATE CONCERNS

2. Data Repository Remains Empty

The revised manuscript cites a new DOI (10.17605/OSF.IO/KWCS8) for data availability. The OSF project exists and is public, but as of 10th March it contains zero files. The project description also still references the original manuscript title.

- I am really sorry about this. OSF has recently overhauled their site, and it is causing me endless grief. The problem has been corrected.

3. Within-Female Slope Interpretation

The within-female slope posterior is tight around zero [-0.06, 0.05], which is informative; it rules out large within-female trade-offs. However, the Discussion's claim that this "reflects genuine biological stability rather than limited statistical power" overstates what the data can show. With $n=2$ clutches per female, each female contributes a single contrast, and the van de Pol and Wright decomposition, while a cleaner parameterisation, does not generate additional information from the paired design. Softening this claim (e.g. "any within-female effect, if present, is small") would better reflect the data.

- Good point. I have used your language to soften this claim.

4. Egg Number Repeatability Contains a Scale Mixing Error

The reported observation scale repeatability for egg number ($R = 2.97 \times 10^{-5}$) is extraordinarily low because the formula mixes scales. The Supplementary Material gives $R = V_F / (V_F + (\mu + \mu^2/\theta))$, where V_F is the random intercept variance. Because the intercept operates on the log scale (Equation 5: $\log(\mu_{ij}) = \beta_0 + \beta_1 \text{Brood} + u_{0i}$), $V_F = \sigma^2_N = 0.078^2 = 0.006$ is a log scale quantity, but the denominator uses observation scale NB variance. Under Nakagawa, Johnson, and Schielzeth (2017), the correct observation scale $V_F = \mu^2 * (\exp(\sigma^2) - 1) \sim 11.3$, yielding $R \sim 0.02$, three orders of magnitude larger.

- Yes, good catch. The revised repeatability is $R = 0.048$ (0, 0.24), which still indicates very low repeatability. Our values (0.05 vs 0.02) likely differ because you used a point estimate in the calculation, whereas I used full posterior propagation. My code accounts for uncertainty in μ , σ^2 , and θ , hence the wide CI.

5. Between-Female Slope Language

The manuscript is appropriately cautious in most places ("weak evidence for divergence," "considerable uncertainty"), but a few phrases imply a clearer result than the posterior supports. The between-female slope is -0.05 with 95% CrI (-0.10, 0.00), so most posterior mass is negative, but the interval does include zero. Phrases such as "the modest among-female trade-off detected here" (Discussion) and "primarily expressed among females" (Abstract) would read more accurately as "suggested" rather than "detected," and "appears to be expressed" rather than "is expressed."

- I have toned the language in these places.

MINOR COMMENTS

1. Figure 1 y-axes lack units. Figure 2 y axes lacks labels.

- Axis titles have been modified/added.

2. The graphical abstract has a typo: "aquisition" should be "acquisition" in the right panel.

- The spelling error has been corrected.

3. Within-clutch egg size variation receives only brief interpretive discussion despite a strong result. A sentence or two connecting this to the resource constraint interpretation (e.g. less-uniform provisioning when resources are depleted) seems warranted.

- I modified the third paragraph in the Discussion to better link the within-female egg size variation finding with the resource-constraint hypothesis.

Reviewer: 2

Recommendation: Major Revision

Comments:

While some aspects were improved, other parts remain obscure (e.g., link to a very similar study on the same species) and an apparent contradiction has now emerged (that there is an among-individual correlation involving a non-repeatable trait). Many revisions seem to have been made without direct links to reviewer comments, and the explanations provided in the rebuttal letter were sometimes unhelpful. In most cases, peer review succeeds in making a manuscript better, but it's not obvious this is the case here.

In the revised version, an apparent contradiction emerges. One of the main new results is that now there seems to be a weak trade-off, in which females that produced more eggs on average laid smaller eggs. However, the repeatability of egg number is zero. This is a contradiction, because there can be no among-female covariation involving one trait that does not vary among females. This can't be left as is without any re-consideration of the model, or further explanation.

- At first glance, the weak negative association between mean egg number and egg size among females may appear inconsistent with the near-zero repeatability of egg number. However, these quantities capture different aspects of variation. Repeatability reflects the proportion of unexplained variance attributable to consistent differences among females, whereas the between-female slope represents a systematic (fixed-effect) relationship between female means. Thus, even when residual among-female variance in egg number is very small on the observation scale, it is still possible to detect a weak association between female mean egg number and egg size. In this case, the near-zero repeatability indicates that most variation in egg number arises within females and from distribution-level variance, while the between-female slope suggests only a modest and highly constrained divergence in average allocation strategy. Accordingly, the evidence for an among-female trade-off should be interpreted as weak and driven by small differences in female means rather than by substantial or repeatable variation in clutch size. Necessary edits have been made to the Results and Discussion sections of the manuscript to improve clarity.

It is deceptive to just cite Koch and Meunier (2014) in the introduction after simply saying that the “pattern of allocation to egg number and size is unclear”. Just reading the abstract, the study in question “found negative associations between egg number and mass as well as between egg number and maternal expenditure on egg care. However, these trade-offs could only be detected after statistically correcting for female weight at egg laying”. This is not “unclear” at all, and these results should be explicitly covered in the introduction.

- I charitably said that the pattern observed by Koch and Meunier (2014) is unclear. Reading past the Abstract shows that Koch and Meunier (2014) reported two contradictory correlations that suffer from pseudoreplication and/or an erroneous control for female body size.

The statistical model was revised, but in a strange way that needs more explanation and justification. An oddity with the new bivariate mixed model is that the variable “egg number” now appears both as a response variable and as a fixed effect fitted to the other variable (twice, as a between and within individual component). The justification for this approach needs more explanation. Here are some questions/comments about the new approach:

- 1) In the methods (line 171), it seems that the goal to include the fixed effect of “within-female component of egg number” was to capture covariance arising from deviations within females across clutches. It remains unclear why a residual covariance could not achieve this.
 - Thank you for this insightful comment. Residual covariance (e.g., via `rescor()`) does not provide an appropriate estimate of within-female covariance in our model for two reasons. First, the two response variables are modelled using different distributions (negative binomial for egg number and Gaussian for egg size), so that residual correlations are not defined consistently across responses in this framework. Second, and more importantly, residual covariance represents unexplained covariance after accounting for predictors and does not isolate within-individual effects. In contrast, the within-between decomposition of van de Pol and Wright (2009) explicitly separates within-female deviations from among-female differences, allowing the within-female component to be interpreted as a slope describing plastic changes in egg size as a function of deviations in egg number relative to a female’s own mean. This parameter directly tests the presence of a within-individual trade-off, which cannot be inferred from residual covariance alone. We have clarified this point in the Methods.
- 2) There seems to be a redundancy between the fixed and random components of the model. The ID random component captures inter-individual covariance. The fixed effect component also captures the female differences in average number of eggs. It is argued in the methods, that the random effect captures for “latent covariance”, whereas the fixed effect component captures a “strategic” association. This is confusing to the naïve reader, and more explanation/definition of the terms latent vs strategic covariance is needed. Also, a justification of this approach should be provided to convince readers that it accomplishes what it intends to do.
- 3) Regarding the point #2 above, the interpretation in the discussion seems to be mixing the two. In the results, it is the fixed effect component that is significant, whereas the random-intercept covariance is not significant. In the discussion (lines 308-311), when interpreting the among-female relationship level, it is stated that this “pattern corresponds to the estimated correlation between female-specific intercepts in the multivariate model”. When going back to the methods, the term “between female-level intercepts” is used to designate the random component, which is not significant. Which is which, then?

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- Thank you for these two helpful comments. I agree that my original wording did not clearly distinguish between the roles of the fixed and random components in the model, leading to confusion. I have revised the Methods and Discussion to clarify that the fixed between-female effect (i.e., the between-individual component of egg number) estimates the systematic, explained association between egg number and egg size among females, and therefore captures differences in mean allocation strategy. In contrast, the correlation between female-level random intercepts represents residual (latent) covariance, that is, covariance among baseline trait values that remains after accounting for the fixed effects, including the between-female relationship. Importantly, the negative among-female association reported in the Results is derived from the fixed between-female slope, not from the random intercept correlation, which was highly uncertain. I have revised the Discussion accordingly to ensure that interpretation is consistently aligned with these model components.

Minor comment

Line 151: is “within-clutch processes” really the correct term here? On line 152 the term “within-female plasticity” is used, which is probably more appropriate to use on line 151 also. Or perhaps “among-clutches within individuals” would be also ok.

- Thank you for this suggestion. I agree that “within-clutch processes” could be misleading given the clutch-level structure of our data. I have revised this to “variation among clutches within females” (and “within-female processes”) to more accurately reflect the level of analysis.

Disentangling within- and between-female egg size–number trade-offs in the European earwig.

Clint D. Kelly^{1,✉}

¹ Département des Sciences biologiques, Université du Québec à Montréal, Montréal, Canada

✉ Correspondence: [Clint D. Kelly <kelly.clint@uqam.ca>](mailto:kelly.clint@uqam.ca)

For Review Only

Abstract

Life-history theory predicts a trade-off between offspring size and number, yet empirical studies often fail to detect the expected negative association. This discrepancy may occur because phenotypic correlations confuse variation in resource acquisition with variation in allocation strategy. Using repeated measures of female European earwigs (*Forficula auricularia*), I partitioned covariance between egg size and egg number into among- and within-female components using Bayesian mixed models. The raw phenotypic correlation was near zero. However, females that produced more eggs on average laid smaller eggs, revealing a modest negative association among females that aligns with an allocation trade-off. In contrast, there was no detectable within-female covariance once clutch order was accounted for. Egg number repeatability was effectively zero, suggesting that clutch size varies mainly within rather than among females, while egg size was highly repeatable. Both egg number and egg size decreased in second clutches, and within-clutch egg-size variance increased, indicating a reproductive constraint (i.e., fewer functional ovarioles) under a possibly diminished resource state. Together, these findings demonstrate that the size–number trade-off is primarily appears to be expressed among females through consistent differences in allocation strategy, whereas clutch size itself remains flexible. By separating hierarchical sources of variation, this study shows that allocation trade-offs can remain hidden at the phenotypic level and that evolutionary constraints act through stable allocation rules rather than through flexible clutch-size adjustments.

Introduction

Life-history theory predicts that organisms must allocate finite resources among competing fitness components. Because energy and time are limited, investment in one trait constrains investment in others (Roff, 1992; Stearns, 1992). A central expression of this principle is the offspring size–number trade-off: for a given reproductive budget, producing larger offspring reduces the number that can be produced (Parker, 1982; Smith and Fretwell, 1974). Since larger eggs often enhance offspring performance and survival (Clutton-Brock, 1991), selection is expected to favour an optimal balance between egg size and egg number that maximises maternal fitness (Fox and Czesak, 2000). Yet empirical tests of the size–number trade-off often fail to reveal the expected negative relationship. Across individuals within populations, egg size and egg number are often positively correlated (Bernardo, 1996; Fox and Czesak, 2000; Hargitai et al., 2005; Roff, 1992). This apparent contradiction arises because phenotypic correlations conflate two distinct biological processes: variation in resource acquisition and variation in allocation strategy. Van Noordwijk and De Jong (1986) showed that when variance in acquisition exceeds variance in allocation, individuals with greater resource pools can produce both more and larger eggs, generating positive among-individual correlations even if all individuals are constrained by the same underlying allocation trade-off. In such cases, the population-level pattern masks allocation constraints operating within individuals—an outcome conceptually analogous to Simpson’s paradox (Simpson, 1951).

From this perspective, the critical prediction of life-history theory is not simply a negative phenotypic correlation, but a negative within-individual covariance between egg size and egg number once among-individual differences in acquisition are accounted for (Careau and Wilson,

2017). Detecting such a masked trade-off, therefore, requires partitioning phenotypic variance and covariance into hierarchical components. Repeated measures of individuals allow estimation of (i) among-individual variance, reflecting consistent differences among females in total reproductive investment, and (ii) within-individual variance, reflecting how females adjust allocation across reproductive bouts. Careau and Wilson (2017) argue that only by decomposing covariance into these components using multivariate mixed models can we determine whether allocation trade-offs operate within individuals despite positive population-level associations. Crucially, this framework also predicts that traits involved in allocation trade-offs should show substantial repeatability. If females differ consistently in resource acquisition, egg size and egg number should exhibit among-individual variance across clutches. High repeatability would, therefore, indicate persistent differences in total reproductive investment among females, consistent with variation in acquisition. ~~At the same time, detecting a negative~~ The marked increase in within-individual covariance would indicate that, conditional on their clutch egg size variance in second broods further supports a resource pool constraint interpretation. Under reduced resource availability, females trade-off egg size against egg number may be less able to maintain uniform provisioning across all eggs, leading to greater heterogeneity in egg size within a clutch. Such increased variance is consistent with diminished allocation precision when reproductive events. Thus, variance partitioning and repeatability resources are not merely statistical descriptors, but tests limited, rather than an adaptive diversification of the underlying acquisition-allocation structure predicted by life-history theory offspring size. Despite the foundational importance of the size–number trade-off, empirical tests rarely partition covariance into within- and among-individual components to distinguish allocation from

acquisition effects. Here, I address this gap using the European earwig, *Forficula auricularia*, a species that allows repeated measures of reproductive investment within females but whose pattern of allocation to egg number and size is unclear (Koch and Meunier, 2014). Female earwigs lay a clutch within a short temporal window in a subterranean burrow (Meunier, 2024) and provide extended maternal care (Meunier, 2024; Staerkle and Kölliker, 2008). In Quebec populations, environmental constraints typically render females effectively semelparous (Tourneur, 2018), but under favourable laboratory conditions, they can produce a second clutch. This system, therefore, enables direct estimation of within-female covariance across successive reproductive bouts. I test three predictions derived from acquisition–allocation theory (Van Noordwijk and Jong, 1986; see also Reznick, Nunney, and Tessier, 2000). First, if variance in resource acquisition among females exceeds variance in allocation strategy, then egg size and egg number should exhibit substantial repeatability and a positive among-individual correlation, whereas the within-individual correlation should be negative, revealing a masked allocation trade-off. Second, if maternal fitness is maximised at a particular offspring size or number (Smith and Fretwell, 1974), then one trait should remain relatively invariant across clutches while the other adjusts between reproductive bouts (Winkler and Wallin, 1987). Third, if reproductive investment is structured differently across bouts, then variance in egg size may differ between first and second clutches at both the among-female and within-clutch levels (e.g. Hargitai et al., 2005), providing further insight into how selection shapes the distribution of maternal investment across reproductive events.

By separating among-female differences in total reproductive investment from within-female allocation decisions, this study tests a fundamental question in life-history theory: do apparent

positive associations between egg size and number reflect relaxed constraints, or do allocation trade-offs persist but remain hidden within individuals? Demonstrating a negative within-female covariance despite positive population-level patterns would show that acquisition can mask, but not erase, the constraints imposed by finite resources. In this way, my study evaluates not only the existence of a trade-off but also the hierarchical structure through which life-history constraints are expressed.

Methods

I collected adult female earwigs (n=42) in September and October 2024 in an urban park (Parc Outremont) in Montréal, Canada. All earwigs were brought into the laboratory, assigned a unique ID, and individually placed in a 90-mm-diameter Petri dish containing a thin layer of moist sand. Females were provided a cotton-plugged water vial and fed TetraMin Tropical fish flakes ad libitum. Females were maintained in an incubator (Percival Scientific, Iowa, USA) at 10°C in constant darkness.

Females were checked daily for oviposition. When a female laid eggs, I recorded the date and collected all eggs from the sand and placed them on filter paper in a new Petri dish. The eggs were then photographed under a Leica S6D stereo microscope (Leica Microsystems Inc., Concord, ON, Canada) at 0.68x magnification using Enersight software (Leica Microsystems Inc., Concord, ON, Canada), which automatically adds a scale bar to the photo. Eggs were then discarded.

To stimulate females to lay a second clutch of eggs, I recreated a brief period of spring where they were allowed to accumulate resources for egg production and replenish their sperm stores.

To achieve this, females that had recently oviposited were placed in a new Petri dish containing a cotton-plugged water vial, TetraMin Tropical fish flakes, and a shelter consisting of four 4 cm-long straws (Blowholes, Markham, Canada) glued together. These Petri dishes were placed in a 23 °C incubator with a 12L:12D light: dark cycle. Females were maintained under these conditions for two weeks, at which time a haphazardly chosen stock male from our colony was placed with her. The pair was kept together for a week, after which a new stock male replaced the original male, and that pair was kept together for another week. Females were then transferred to the 10 °C incubator and treated as previously until they laid their second clutch, at which point their eggs were photographed. Females were euthanised by freezing after laying their second clutch, and their pronotum was photographed as described above. Because female fecundity is often related to body size in insects (Honek, 1993) and could thus possibly explain a positive among-female covariance between egg number and size, I measured two proxies of size, pronotum length and body mass. Females were weighed to the nearest 0.01 mg using a Sartorius Secura 224-1S analytical balance. Females were weighed three times, with each weighing bout separated by four days. I used Fiji image analysis software (Schindelin et al., 2012) to measure the length of the female pronotum to the nearest 0.001 mm three times, with each measurement separated by 4 days. I used the same procedure to measure the perimeter of each egg.

131 **Statistical analysis**

I calculated the repeatability of the body mass, pronotum length and egg perimeter measurements using the R package *rptR* (Stoffel, Nakagawa, and Schielzeth, 2017). I entered the trait of interest as the response variable, female ID as a random effect, and assumed a Gaussian distribution. Each analysis was based on 1000 bootstrap samples. The measurements of my two

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4 136 morphological traits were highly repeatable (pronotum length: R [95% CI] = 0.98 [0.97, 0.99];
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6 137 body mass: R [95% CI] = 0.99 [0.988, 0.996]); therefore, I used each female's mean value for
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8 138 each trait where necessary. Because egg perimeter was highly repeatable in both clutches (first
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10 139 clutch: R [95% CI] = 0.977 [0.976, 0.979]; second clutch: R [95% CI] = 0.93 [0.926, 0.937]), I
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12 140 calculated each egg's average perimeter and then used these values to calculate the mean egg size
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14 141 for each clutch for each female.
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18 142 To reduce the two correlated measures of body size (pronotum length and body mass) to a single
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20 143 structural size metric, I performed a principal components analysis (PCA) on the correlation
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22 144 matrix of standardized variables using the *prcomp* function in the R stats package (R Core Team,
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24 145 2025). The first principal component (PC1) explained 60.8% of the total variance and was used
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26 146 as an index of structural body size. In addition, female body condition was estimated using the
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28 147 scaled mass index (SMI) following Peig and Green (2010, 2009; see also Kelly, Tawes, and
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30 148 Worthington, 2014), which standardizes body mass relative to structural size (pronotum length).
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35 149 I first quantified associations among reproductive traits using raw phenotypic correlations.
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37 150 Pearson correlation coefficients were calculated using individual-level trait values, thereby
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39 151 capturing total phenotypic covariation across all sources of variation. These correlations reflect
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41 152 the combined contribution of consistent differences among females, ~~within-clutch~~-variation
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43 153 among clutches within females, and residual effects. Consequently, raw phenotypic correlations
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45 154 provide a descriptive estimate of overall trait associations but do not distinguish between
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47 155 covariation arising from stable female differences and that arising from within-female (i.e.,
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49 156 among-clutch) processes.
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To distinguish covariance arising from within-female plasticity from covariance attributable to consistent among-female differences, I followed the within–between decomposition approach of van de Pol and Wright [(2009); see Supplementary Material for details]. This decomposition is necessary because residual covariance does not isolate within-individual effects and, in models with different response distributions, does not provide a directly interpretable estimate of within-female trade-offs. Although mean-centering has been shown to impose assumptions about underlying biological processes (Westneat et al., 2020), these assumptions are appropriate here because the biological question concerns how reproductive allocation varies within females across clutches versus how females differ consistently in allocation strategy. Egg number was decomposed into two components: a between-female component, defined as each female’s mean egg number across broods centered on the population mean, and a within-female component, defined as the deviation of each brood from that female’s own mean. Hereafter, we refer to “among-female” variation as differences between individuals and “within-female” variation as differences among clutches within individuals. These two predictors were included simultaneously in a Bayesian multivariate mixed-effects model in which egg number and mean egg size were modelled jointly. Egg number was modelled using a negative binomial distribution with a log link to account for overdispersion, whereas mean egg size was modelled assuming a Gaussian distribution with an identity link. Clutch (first vs. second clutch) was included as a fixed effect in both submodels. Female identity was included as a random intercept for both response variables. This structure allowed estimation of among-female variance in egg number, among-female variance in egg size, and the covariance (correlation) between female-level intercepts for the two traits.

This multivariate framework partitions total phenotypic covariance into ~~two components:~~
~~covariance~~ components arising from consistent differences among females and from within-
female deviations across clutches. Covariance attributable to consistent differences among
 females is captured by the female-level random effects, whereas covariance arising from
~~deviations within females across clutches~~ female variation is captured by the fixed within-female
 component of egg number. The fixed between-female component ~~quantifies~~ estimates how mean
 egg size ~~differs~~ varies with mean egg number across females, and thus captures systematic (i.e.,
explained) differences in allocation strategy among females ~~based on their average egg number,~~
~~thereby estimating the among-female (strategic) association.~~ In contrast, the fixed within-female
 component ~~estimates~~ tests whether egg size changes plastically when a female produces more or
 fewer eggs than her own mean. The correlation between female-level intercepts represents
residual (latent) covariance ~~in~~ among baseline trait values ~~among females after accounting for~~
~~all that is not explained by the~~ fixed effects, including the between-female allocation relationship.
 Thus, the between-female slope tests whether females diverge in their mean allocation strategy
 (i.e., whether females that consistently produce more eggs also ~~consistently~~ produce smaller
 eggs). ~~In contrast,~~ whereas the intercept correlation ~~tests for latent~~ quantifies any remaining
 covariance between ~~baseline~~ egg number and egg size after accounting for ~~brood~~ these effects ~~and~~
~~the within-between decomposition, reflecting residual integration not explained by mean~~
~~allocation differences.~~ Continuous predictors were standardized prior to analysis. ~~Standardized,~~
such that standardized coefficients, ~~therefore,~~ represent the expected change in egg size,
~~measured~~ (in standard deviation units,) associated with a one standard deviation change in egg
 number at the relevant hierarchical level.

To quantify the structure of egg-size variation specifically, I fitted a univariate mixed-effects model in which egg size (standardised mean egg perimeter) was modelled as a Gaussian response with female identity included as a random intercept (see Supplementary Material for details).

This formulation partitions total phenotypic variance in egg size into variance attributable to differences among females and variance occurring within females among eggs within clutches.

The among-female variance component reflects consistent maternal differences in egg size allocation, whereas the within-female component captures heterogeneity in provisioning within clutches and residual variation.

Repeatability was calculated on the observation scale for both traits. For egg size (Gaussian model):

$$R_{\text{egg size}} = \frac{V_F}{V_F + V_R}$$

where V_F is the among-female variance (i.e., variance of the female random intercept) and V_R is the residual variance. Because the Gaussian model is parameterized directly on the observation scale, this formulation corresponds to both latent- and observation-scale repeatability.

For egg number, which was modelled using a negative binomial distribution with a log link, the variance depends on both the expected mean and the dispersion (shape) parameter. Under the negative binomial parameterization used in *brms*, the observation-level (distribution-specific) variance is given by:

$$\text{Var}(Y) = \mu + \frac{\mu^2}{\theta}$$

where μ is the expected mean egg number and θ is the dispersion (shape) parameter.

~~Observation-scale repeatability for egg number~~Because the model includes a log link, the among-female variance estimated from the random intercept is expressed on the latent (log) scale and must be transformed to the observation scale before calculating repeatability. Following Nakagawa et al. (2017), the among-female variance on the observation scale was therefore calculated as:

$$V_F^{obs} \equiv \mu^2 (\exp(\sigma^2) - 1)$$

where σ^2 is the variance of the random intercept on the latent scale.

~~Observation-scale repeatability for egg number~~ was then calculated as:

$$R_{\text{egg number}} = \frac{V_F}{V_F + \left(\mu + \frac{\mu^2}{\theta} \right)} \frac{V_F^{obs}}{V_F^{obs} + \left(\mu + \frac{\mu^2}{\theta} \right)}$$

~~where V_F is~~Both the ~~transformed~~ among-female variance ~~estimated from the random intercept.~~ ~~The and the~~ distribution-specific variance ~~term $\mu + \mu^2/\theta$ was calculated~~ were computed for each posterior draw using the corresponding posterior samples of μ and θ , ~~thereby generating σ^2 , and θ .~~ To account for variation in expected values across observations, these quantities were first calculated at the observation level and then averaged within each posterior draw. This procedure yielded a posterior distribution of repeatability– ~~that fully propagates uncertainty from all variance components.~~

Repeatability for non-Gaussian traits was calculated following the framework of Nakagawa and Schielzeth (2010) and ~~the~~its extensions for generalized linear mixed models ~~described by~~

(Nakagawa ~~et al.~~, [Johnson, and Schielzeth, 2017](#)). Because distribution-specific variance can contribute substantially to total phenotypic variance in negative binomial models, repeatability estimates for count data are often lower than those obtained under Gaussian assumptions, even when among-individual variance is present (Nakagawa, Johnson, and Schielzeth, 2017; Nakagawa and Schielzeth, 2010). By calculating repeatability on the observation scale ~~for both traits~~, these estimates represent the proportion of total phenotypic variance in ~~the~~ observed ~~trait~~ egg number attributable to consistent differences among females ~~(Nakagawa and Schielzeth, 2010). Derived repeatability values were calculated for each posterior draw to propagate uncertainty from all variance components (Villemereuil, Gimenez, and Doligez, 2013).~~

I used weakly informative priors to constrain parameters to biologically realistic ranges while allowing the data to drive inference. Regression coefficients were assigned normal priors centered at zero ($\text{Normal}(0, 0.5-1)$), reflecting the expectation that clutch effects on egg traits would be moderate in magnitude. Intercepts were given normal priors scaled to each trait (egg number: $\text{Normal}(0, 5)$; mean egg size: $\text{Normal}(3.5, 1)$; standardized egg size: $\text{Normal}(0, 0.5)$). Among-female variation was assigned exponential priors ($\text{Exponential}(1-2)$), which favour small but non-zero individual differences. Residual variance (Gaussian models) and the negative binomial dispersion parameter were given $\text{Exponential}(1)$ priors. Correlations among female-level random effects were regularized using an $\text{LKJ}(2)$ prior to avoid overestimating trait associations. Posterior distributions were obtained using six Markov chains with 4,000 iterations per chain, including 1,000 warm-up iterations, and convergence was assessed using $\hat{R}$ statistics, effective sample sizes, and trace-plot inspection. Posterior distributions are summarised as

posterior means with 95% credible intervals. Data were visualized using *ggplot* (Wickham, 2016) and *tidybayes* (Kay, 2020).

Results

Between- and within-female associations

Across all broods, the raw phenotypic correlation between egg number and egg size was weak and near zero ($r_{raw} = -0.04$, 95% CI [-0.25, 0.19]), indicating no detectable population-level association when hierarchical structure was ignored.

Egg number (clutch one: $\bar{x} = 51.55 \pm 2.49$ eggs; clutch two: $\bar{x} = 34.14 \pm 2.65$ eggs) and egg size (first clutch: $\bar{x} = 3.54 \pm 0.03$ mm; second clutch: $\bar{x} \pm SE = 3.41 \pm 0.03$ mm) both declined in second clutches. Egg number decreased on the log scale ($\beta = -0.41$, 95% CrI (-0.64, -0.19; Figure 1; Supplementary Table 1), corresponding to an approximate 34% reduction on the natural scale ($e^{-0.413} \approx 0.66$). Mean egg size also declined in the second clutch ($\beta = -0.13$, 95% CrI (-0.24, -0.02)).

The results did not support my prediction of a within-female plastic trade-off between egg number and egg size (Figure 1 a). The within-female deviation in egg number had a slope near zero ($\beta_W = -0.01$, 95% CrI (-0.06, 0.05), indicating that females did not systematically adjust egg size when producing more or fewer eggs relative to their own mean. ~~In contrast~~By comparison, the ~~among-female association (captured by the fixed~~ between-female ~~componentslope)~~ revealed a negative ~~associationrelationship~~ between egg size and mean female egg number ($\beta_B = -0.05$, 95% CrI (-0.10, 0.00; Figure 1 b). The posterior distribution of the among-female slope was centered on negative values, suggesting weak evidence for divergence in ~~mean~~ allocation strategy among

females. The correlation between female-level intercepts for egg number and egg size was highly uncertain (median = -0.02, 95% CrI: -0.81, 0.82), indicating limited information for estimating latent covariance with only two broods per female. With only two repeated measures per female, intercept correlations are weakly identifiable and should be interpreted cautiously.

No measure of female size/condition was positively correlated with the number of eggs in the first brood (pronotum length: $r = 0.14$, 95% CI [-0.19, 0.43]; body mass: $r = 0.12$, 95% CI [-0.21, 0.43]; PC1: $r = -0.17$, 95% CI [-0.44, 0.15]; scaled mass index: $r_{raw} = -0.02$, 95% CI [-0.35, 0.33]) providing no evidence that larger females or those in better condition produce more eggs. I therefore did not include a measure of size/condition in this model. Together, these results show that the size–number trade-off is not apparent at the raw phenotypic level and is not expressed as within-female plasticity, but emerges as a modest negative association at the among-female level through differences in average allocation strategy.

Egg size variation between females and within clutches

Egg size exhibited substantial variation at both the between-female and within-clutch levels (Figure 2; Supplementary Table 2). Between-female variance was similar across reproductive events, with a median of 0.73 (95% CrI: 0.50, 1.12) in first clutches and 0.61 (95% CrI: 0.41, 0.96) in second clutches. The posterior median difference (second – first) was -0.11 (95% CrI: -0.56, 0.28), and the posterior probability that between-female variance was greater in second clutches was 0.28, indicating no clear evidence of a difference. In contrast, within-clutch (egg-to-egg) variance was substantially higher in second clutches (median = 0.35 (95% CrI: 0.33, 0.38) than in first clutches (median = 0.21 (95% CrI: 0.20, 0.22). The posterior median difference was

0.14 (95% CrI: 0.12, 0.17), with a posterior probability of 1.00, that within-clutch variance was greater in second clutches.

Egg number and egg size repeatability

Repeatability was calculated from posterior samples of the variance components on the observation scale. Egg number exhibited extremely low repeatability. Although there was some among-female variance in egg number (SD = 0.078, 95% CrI [0.003, 0.270]; Supplementary Table 1), the overwhelming majority of variance arose from distribution-specific (negative binomial) variance. Consequently, observation-scale repeatability was near zero ($R = 2.974.86 \times 10^{-5} 10^{-2}$, 95% CrI [2.01×10^{-8} , 1.61×10^{-4}] [3.80×10^{-5} , 2.43×10^{-1}]), indicating that very little of the total phenotypic variation in clutch size was attributable to consistent differences among females.

In contrast, egg size showed moderate repeatability on the observation scale. Among-female variance (SD = 0.082, 95% CrI [0.006, 0.155]; Supplementary Table 21) was substantial relative to residual variance (SD = 0.191, 95% CrI [0.155, 0.235]), yielding $R = 0.78$ (95% CrI: 0.70, 0.84) in first broods and $R = 0.63$ (95% CrI: 0.53, 0.73) in first broods and $R = 0.78$ (95% CrI: 0.70, 0.84) in second broods. These results indicate that females differed consistently in their mean egg size across broods, whereas clutch size was largely context-dependent.

Discussion

Life-history theory predicts a trade-off between offspring size and number because both traits draw from a finite pool of reproductive resources (Lack, 1947; Roff, 1992; Smith and Fretwell, 1974; Stearns, 1992). Yet, as emphasised by van Noordwijk and de Jong (1992), phenotypic

correlations need not directly reveal allocation trade-offs, because variation in resource acquisition can obscure underlying allocation rules (Careau and Wilson, 2017; Johnson and Nassrullah, 2024). This acquisition–allocation framework has become central to interpreting life-history covariance (Houle, 1991; Reznick, Nunney, and Tessier, 2000). By partitioning covariance into within- and among-female components, my results show that the size–number trade-off in this population of earwigs is ~~modestly~~at most weakly expressed among females ~~but~~and absent within them, illustrating how hierarchical structure shapes the detectability of life-history trade-offs (Careau and Wilson, 2017; Dingemanse and Dochtermann, 2013; Pol and Wright, 2009; Westneat et al., 2020).

At the phenotypic level, egg size and egg number were uncorrelated in my earwig population, consistent with Koch and Meunier (2014). Under the framework of van Noordwijk and de Jong (1986), such phenotypic independence is expected when variation in resource acquisition obscures underlying allocation trade-offs. After hierarchical decomposition, ~~however~~, a weak negative association emerged at the among-female level: females that produced more eggs on average tended to lay slightly smaller eggs. This pattern ~~corresponds to~~is captured by the ~~estimated fixed between-female slope and reflects systematic differences among females in their mean allocation strategy. In contrast, the~~ correlation between female-~~specific~~level intercepts ~~in the multivariate model and suggests modest divergence in average allocation strategy among females. Nevertheless, the credible interval surrounding—representing residual (latent) covariance after accounting for this intercept correlation—allocation relationship—was highly uncertain and~~ overlapped zero, indicating ~~considerable uncertainty in the strength of among-female covariance. Thus, rather than forming a tightly integrated life-history syndrome, egg size~~

~~and number appear to vary along a shallow axis of allocation divergence embedded within~~
~~substantial variation in resource acquisition~~ little evidence for additional integration between traits
~~beyond that explained by differences in mean allocation strategy~~ (Roff and Fairbairn, 2007;
Stearns, 1989).

Crucially, these results must be interpreted in light of the variance structure of egg number. Egg
number exhibited near-zero repeatability, indicating that there is very little residual among-
female variation in clutch size once distribution-level variance is taken into account. At first
glance, this may appear inconsistent with the presence of an among-female association. However,
repeatability and the between-female slope quantify different components of variation.
Repeatability reflects the proportion of unexplained (random-effect) variance attributable to
consistent differences among females, whereas the between-female slope represents a systematic
(fixed-effect) relationship among female means. Thus, even when residual among-female
variance is very small, it is still possible to detect a weak association among female means. In
this case, the near-zero repeatability indicates that clutch size does not vary consistently among
females, while the between-female slope suggests only minor differences in average reproductive
output that covary weakly with egg size. Taken together, these results indicate that the apparent
among-female trade-off is weak and arises from small differences in female means rather than
from substantial or repeatable variation in clutch size. Accordingly, there is limited evidence for
stable divergence in allocation strategy among females.

In contrast, there was no evidence for a within-female plastic trade-off. Although each female
contributed only two broods, the posterior distribution of the within-female slope was tightly
centered on zero, suggesting that ~~this absence reflects genuine biological stability rather than~~

~~limited statistical power~~any within-female effect, if present, is small. Females did not reduce egg size when producing more eggs than their own average; instead, egg number and egg size shifted together across broods. This pattern is consistent with state-dependent life-history models in which allocation rules remain relatively stable while total resource state fluctuates across reproductive bouts (Houston and McNamara, 1999; McNamara and Houston, 1996). Rather than dynamically reallocating resources between offspring size and number, females appear to exhibit clutch-level shifts driven by overall energetic state. Together, these results suggest that clutch size is primarily state-dependent and influenced by transient physiological or structural constraints, whereas egg size reflects a more stable component of reproductive allocation.

Notably, egg number was not correlated with female structural size or body condition (i.e. scaled mass), indicating that persistent differences in morphology do not explain clutch-size variation. If stable acquisition differences were driving among-female variation, we would expect egg number to show repeatability and to covary with these morphological size/condition indicators. The absence of such patterns weakens the interpretation that persistent acquisition differences—at least as reflected by morphology—underlie among-female differences in reproductive output. Instead, clutch size appears predominantly state-sensitive and influenced by clutch-level constraints.

This interpretation also helps explain the pronounced decline in second clutches. That second clutches comprised both fewer and smaller eggs is consistent with previous work in North American and European populations (Koch and Meunier, 2014; Tourneur and Gingras, 1992). Tourneur and Gingras (1992) proposed that some ovarioles become non-functional after the first clutch, thereby reducing egg production capacity. Structural limitation could therefore contribute

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4 389 to smaller second clutches. However, structural reduction alone cannot explain the simultaneous
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6 390 decline in egg size. Under a strict allocation trade-off, a reduction in egg number would be
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8 391 expected to lead to compensatory increases in egg size if total resources remain constant. Instead,
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10 392 both traits declined, indicating that females likely entered the second reproductive bout with
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12 393 diminished resource reserves in addition to any physiological constraint. In the wild, females
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14 394 accumulate resources for months before producing their first clutch, whereas in the laboratory,
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16 395 they had only a brief interval (approx. one month) to replenish reserves before producing the
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18 396 second. Late reproduction, therefore, likely reflects both reduced energetic state and structural
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25 398 The hierarchical variance structure supports a constraint-based interpretation of clutch-size
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27 399 variation. Egg size exhibited moderate repeatability across broods, indicating stable differences
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29 400 among females in mean per-offspring investment. In contrast, egg number showed extremely low
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31 401 repeatability, with most variance arising from brood effects and distribution-level variance rather
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33 402 than persistent female differences. The absence of any association between egg number and
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35 403 female structural size or condition, together with the near-zero repeatability of clutch size,
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37 404 suggests that variation in egg number is not driven by stable differences in resource acquisition
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39 405 but instead reflects transient physiological state or clutch-level constraint. Thus, clutch size
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41 406 appears primarily state-sensitive, whereas egg size reflects a more stable allocation strategy
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43 407 (Houston and McNamara, 1999; McNamara and Houston, 1996). At the same time, within-clutch
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45 408 variance increased substantially in second broods, indicating reduced developmental precision
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47 409 under diminished resource availability. Structural and energetic constraints, therefore, appear to
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reduce total reproductive output while leaving mean allocation rules largely intact (Ricklefs and Wikelski, 2002; Zera and Harshman, 2001).

These patterns have broader implications for phenotypic integration and evolutionary potential. Repeatability places an upper bound on heritability (Falconer and MacKay, 1996; Lessells and Boag, 1987; Nakagawa and Schielzeth, 2013; Nakagawa and Schielzeth, 2010; but see Dohm, 2002). The moderate repeatability of egg size suggests that selection could act on persistent differences in mean provisioning, whereas the extremely low repeatability of egg number implies that clutch size may respond primarily to short-term environmental or physiological conditions (i.e. reduced ovariole function). The weak intercept correlation further indicates that egg size and egg number are only loosely integrated, consistent with models in which traits differ in their sensitivity to resource state versus allocation strategy (Houle, 1991). In this system, per-offspring investment appears more evolutionarily stable than total offspring number.

Viewed hierarchically, my results illustrate a broader conceptual point: life-history trade-offs may operate primarily through ~~stable~~systematic differences in mean allocation ~~rules~~ among individuals, while most short-term variation reflects state-dependent constraint. However, when among-individual variance in one trait is minimal—as is the case for egg number here—such trade-offs will necessarily be weak and difficult to detect. Without decomposing within- and among-individual covariance, ~~the~~even this modest ~~among-female trade-off detected here~~pattern would have remained obscured. More generally, the evolutionary consequences of trade-offs depend not only on their sign but on their hierarchical expression and repeatability. A trade-off that reflects consistent among-individual differences in a repeatable trait is more likely to shape evolutionary responses than one arising primarily from transient, plastic variation within

individuals. Future work measuring resource acquisition directly, experimentally manipulating body condition, or estimating genetic correlations between egg size and egg number will help determine whether the limited allocation divergence observed here reflects adaptive differentiation ~~in allocation strategy~~ or condition-dependent constraint (Lande, 1980; Roff, 1992).

Acknowledgements

I thank Andréanne Nault and Clara Smythe for assistance in animal care and data collection. I also thank Denis Réale, Pierre-Olivier Montiglio, and two anonymous reviewers for their discussion and feedback.

Data availability

~~Data are available through the OSF repository at 10.17605/OSF.IO/KWCS8.~~

Author contributions

~~CDK collected and analyzed the data, and wrote the manuscript.~~

Funding

The study was funded by a Discovery Grant from the Natural Sciences and Engineering Research Council of Canada (NSERC).

Conflict of Interest

I declare no conflict of interest.

Data availability

Data are available through the OSF repository at [10.17605/OSF.IO/KWCS8](https://doi.org/10.17605/OSF.IO/KWCS8).

Author contributions

CDK collected and analyzed the data, and wrote the manuscript.

Ethics Statement

All data were collected using the European earwig (*Forficula auricularia*), an invertebrate species for which institutional animal care approval and collection permits are not required under current regulations. All procedures adhered to standard ethical guidelines for the care and use of animals in research.

Artificial intelligence (AI)-assisted technologies

Alt text for the two figures was generated with the assistance of AI tools and reviewed and verified by the author for accuracy and completeness.

Acknowledgements

~~I thank Andréanne Nault and Clara Smythe for assistance in animal care and data collection. I also thank Denis Réale and Pierre-Olivier Montiglio for their discussion and feedback.~~

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583 ##Figures

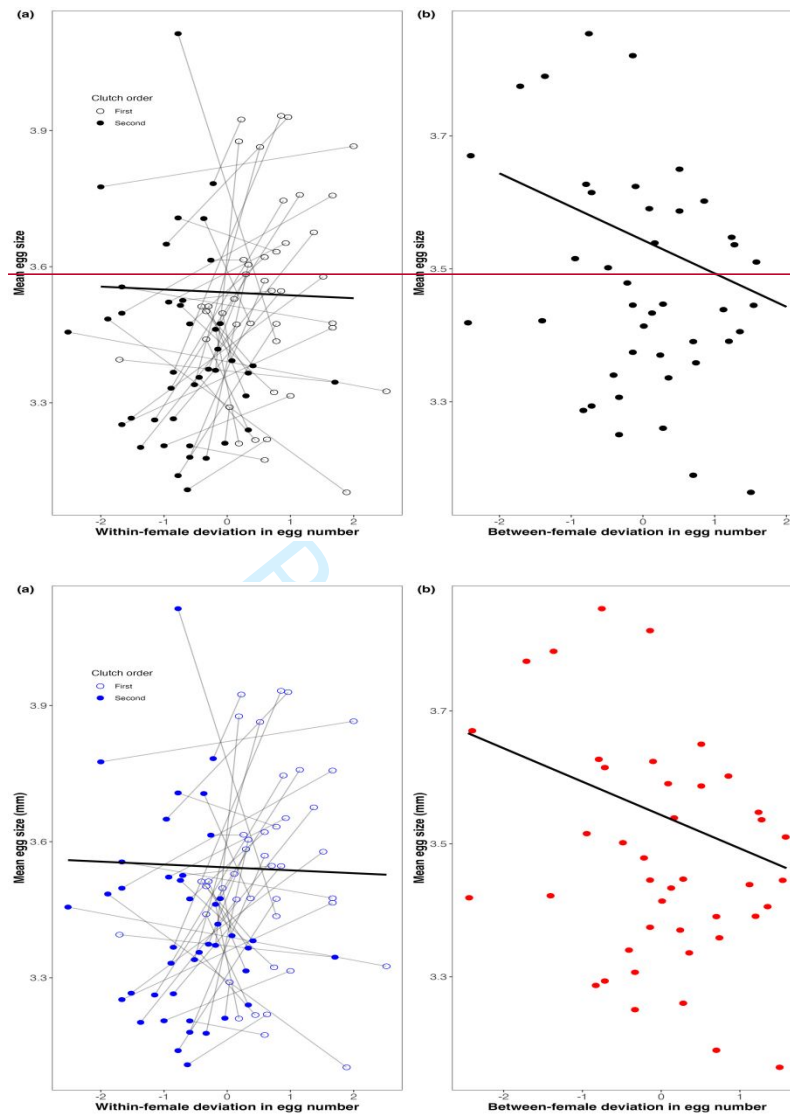


Figure 1: Partitioning the egg size–number trade-off into within- and between-female components. (a) Within-female (plastic) relationship between egg number and mean egg size. Points represent individual broods (first clutch: open circles; second clutch: closed circles), with grey lines connecting repeated measures from the same female. The x-axis shows standardized within-female deviation in egg number (0 = female mean). The solid black line represents the Bayesian population-level estimate of the within-female slope ($\beta_W = -0.01$, 95% CrI (-0.06, 0.05)). (b) Between-female (strategic) relationship (i.e., differences in mean allocation strategy) based on female means across broods plotted against standardized between-female deviation in egg number. The black line represents the Bayesian population-

level between-female slope ($\beta_B = -0.05$, 95% CrI (-0.10, 0.00)). ~~Together, the~~The panels distinguish plastic allocation within females from strategic differences among females, indicating weak support for a reproductive size–number trade-off.

Alt text: Two-panel scatterplot showing relationships between egg number and mean egg size.
Panel (a) plots within-female deviations, with paired points for first and second clutches
connected by lines; the trend line is nearly flat, indicating little within-female trade-off. Panel (b)
plots between-female deviations, with a downward-sloping regression line indicating that females
producing more eggs tend to have smaller eggs.

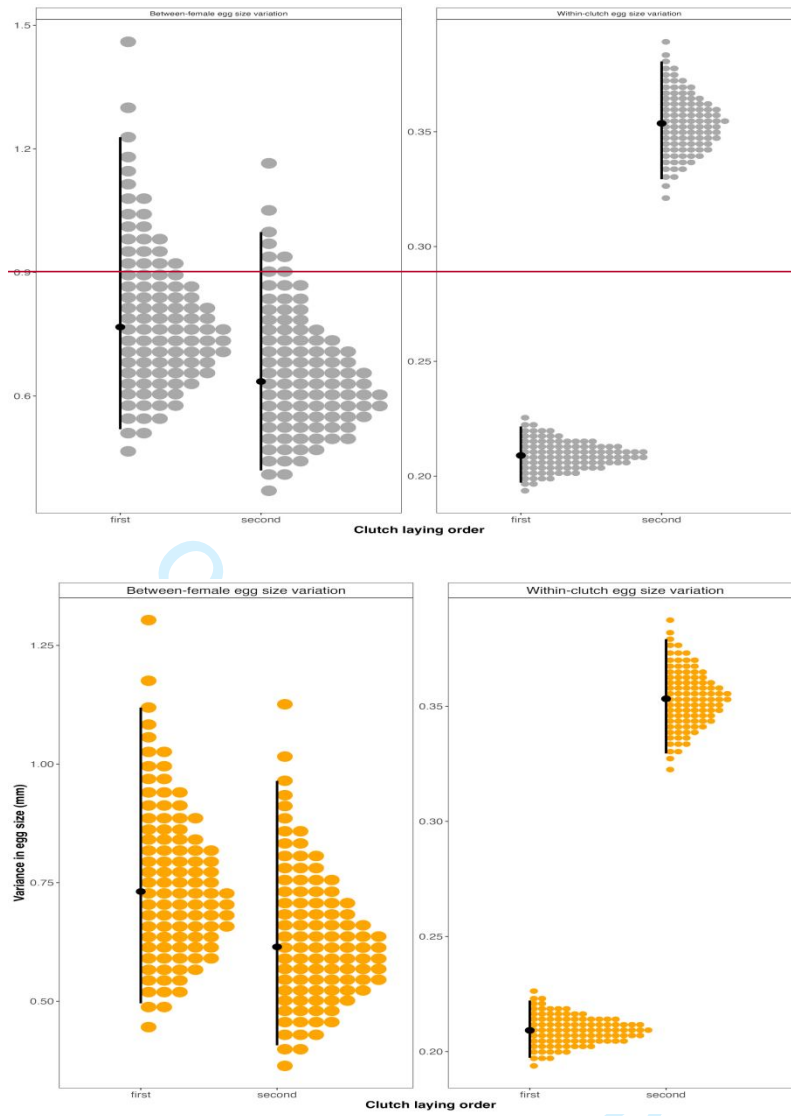


Figure 2: Clutch-specific variance components of egg size. (a) Between-female variance in egg size for first and second clutches, estimated from a hierarchical Bayesian model with clutch-specific female effects. (b) Within-clutch (egg-to-egg) variance for first and second clutches. GreyOrange points represent posterior draws, black points indicate posterior medians, and vertical lines denote 95% credible intervals. Between-female variance did not differ clearly between clutches, whereas within-clutch variance was greater in second clutches.

Alt text: Two-panel dot plot showing variation in egg size across clutch order. The left panel shows between-female variation, with distributions for first and second clutches; variance is higher for first clutches than second clutches, with mean values indicated by black points and

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594 vertical error bars. The right panel shows within-clutch variation, with second clutches exhibiting
595 greater variance than first clutches; means and error bars are shown in black.

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